Friend or Foe:
Delegating to an AI whose Alignment is Unknown

Drew Fudenberg
(MIT)

Annie Liang
(Northwestern/OpenAI)

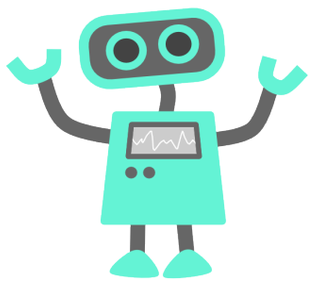
Yale Alignment Workshop

Introduction

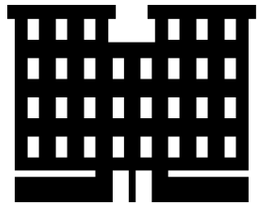


Introduction

Aligned? Misaligned?



AI



FRONTIER LAB

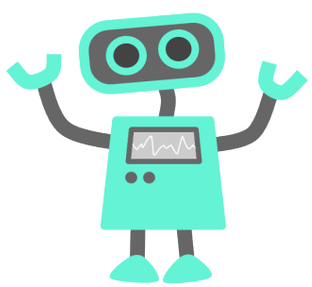
DOCTOR



PATIENT

Introduction

Aligned? Misaligned?



AI

DOCTOR



PATIENT

- The AI possibly knows more than the doctor and can make better decisions.
- But it is also possibly misaligned.
- How can we capture the benefits from superintelligent AI while limiting harms from misalignment, and what are the necessary tradeoffs?
- **Our paper:** simplest possible setting: AI chooses which patients should receive a risky treatment.

Introduction

In our model, the doctor knows treatment success rates by pre-defined groups.

Group	Probability of treatment success
18-30	40%
31-65	20%
66+	10%

The doctor delegates decisions to an AI and disciplines it with two choices:

- A treatment rate per group (e.g., "treat 40% of patients in age group 18-30" or "treat no one in age group 18-30")
- How much is revealed about the patients (e.g., could reveal just the age group or give the AI rich covariates)

How should the doctor make these choices, and what is the risk-reward tradeoff within those groups?

Model

Model

- A designer chooses which patients in the population should receive a risky treatment
  - $a = 1$ is a decision to treat and $a = 0$ is a decision not to treat

- Each patient has a type $y \in \mathcal{Y} = \{0,1\}$—they are either **helped** $(y = 1)$ or **harmed** $(y = 0)$ by treatment

- The designer's payoffs $u(a, y)$ are:
  - 1 from treating someone who is helped by treatment
  - -1 from treating someone who is harmed by treatment
  - 0 otherwise

What the Designer Knows

- G is a group (e.g., the patient's age and gender)
- The designer knows the joint distribution of (G, Y)
  - Let $p_g := P(Y = 1 \mid G = g)$ denote the baseline treatment success rate in group g
- Could simply take the best action given this information
- But hopes to **do better** by giving an AI access to additional inputs X and letting it decide based on (G, X)

How the Designer Delegates

The designer has two levers for reducing the risk of delegation:

1. Committing to a fixed treatment rate r_g per group g
   - Thus the designer decides **how many** patients are treated, while the AI decides **which ones** fill this rate
2. Restricting the informativeness of the AI's inputs X (more on next slide)
   - Limits how well the AI **can identify** patients who do/do not benefit from treatment

Restricting the Inputs

- The designer chooses auxiliary inputs X to give to the AI
- The designer **does not** know the joint distribution of (G, X, Y)
  - Or else could use this information themselves!
- But the designer perceives bounds on how informative X could be
- Specifically, there are bounds $\tau = (\underline{\tau}_g, \overline{\tau}_g)_{g \in \mathscr{G}}$ such that the designer believes

$$P(Y = 1 \mid G = g, X = x) \in [\underline{\tau}_g, \overline{\tau}_g] \quad \forall g \in \mathscr{G}, x \in \mathscr{X}$$

- The designer's **ambiguity set** $\mathscr{P}(\tau)$ consists of all distributions P that are consistent with these bounds and with the known distribution of (G, Y)

Restricting the Inputs

- The designer chooses auxiliary inputs X to give to the AI
- The designer **does not** know the joint distribution of (G, X, Y)
  - Or else could use this information themselves!
- But the designer perceives [illegible] uld be
- Specifically, there are boun[illegible] designer believes

Inputs are unrestricted, could be perfectly informative

$$P(Y = 1 \mid G = g, X = x) \in [0,1] \quad \forall g \in \mathcal{G}, x \in \mathcal{X}$$

- The designer's **ambiguity set** $\mathcal{P}(\tau)$ consists of all distributions P that are consistent with these bounds and with the known distribution of (G, Y)

Restricting the Inputs

- The designer chooses auxiliary inputs X to give to the AI
- The designer **does not** know the joint distribution of (G, X, Y)
  - Or else could use this information themselves!
- But the designer perceives [illegible] uld be
- Specifically, there are boun[illegible] designer believes

X is uninformative about Y for group g

$$P(Y = 1 \mid G = g, X = x) \in [p_g, p_g] \; \forall g \in \mathcal{G}, x \in \mathcal{X}$$

- The designer's **ambiguity set** $\mathcal{P}(\tau)$ consists of all distributions P that are consistent with these bounds and with the known distribution of (G, Y)

Timing

1. The designer commits to treatment rates $r = (r_g)_{g \in \mathcal{G}}$ and conditional success probability bounds $\tau = (\underline{\tau}_g, \overline{\tau}_g)_{g \in \mathcal{G}}$
2. Nature selects a true distribution $P^* \in \mathcal{P}(\tau)$
3. The AI (knowing P^*) selects an allocation rule $a : \mathcal{G} \times \mathcal{X} \to [0,1]$ that is **admissible**, i.e., yields treatment rates consistent with r
4. The designer treats according to a

Timing

1. The designer commits to treatment rates $r = (r_g)_{g \in \mathcal{G}}$ and conditional success probability bounds $\tau = (\underline{\tau}_g, \overline{\tau}_g)_{g \in \mathcal{G}}$
2. Nature selects a true distribution $P^* \in \mathcal{P}(\tau)$
3. The AI (knowing P^*) selects an allocation rule $a : \mathcal{G} \times \mathcal{X} \to [0,1]$ that is **admissible**, i.e., yields treatment rates consistent with r
4. The designer treats according to a

Let $U(P^*, a) = \mathbb{E}_{P^*,a}[u(A, Y)]$ denote the designer's expected payoff when Nature chooses true distribution P^* and the AI chooses allocation rule a.

Pareto Frontier of Best-Case and Worst-Case Payoffs

Worst-case payoff:
$$\underline{v}(r,\tau) = \inf_{P^* \in \mathscr{P}(\tau)} \inf_{a \in \mathscr{A}(r)} U(P^*, a)$$

Best-case payoff:
$$\overline{v}(r,\tau) = \sup_{P^* \in \mathscr{P}(\tau)} \sup_{a \in \mathscr{A}(r)} U(P^*, a)$$

Pareto Frontier of Best-Case and Worst-Case Payoffs

Worst-case payoff: $$\underline{v}(r,\tau) = \inf_{P^* \in \mathscr{P}(\tau)} \inf_{a \in \mathscr{A}(r)} U(P^*, a)$$

Best-case payoff: $$\overline{v}(r,\tau) = \sup_{P^* \in \mathscr{P}(\tau)} \sup_{a \in \mathscr{A}(r)} U(P^*, a)$$

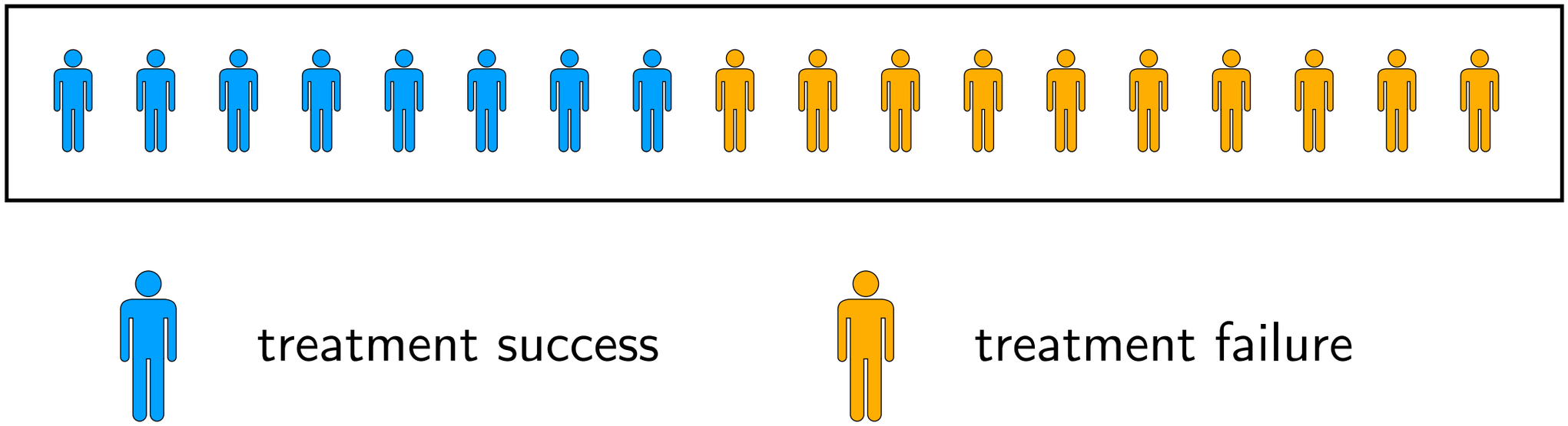
Definition: The **feasible set** $\mathscr{F}$ is the convex hull of all $(\underline{v}, \overline{v})$ that are implemented by some (r, τ), and the **risk-reward frontier** $\mathscr{E}$ consists of all $(\underline{v}, \overline{v}) \in \mathscr{F}$ that are undominated in $\mathscr{F}$ (in the usual Pareto order)

Can trace out this frontier by considering designers who linearly aggregate the worst-case and best-case payoffs: $\eta \cdot \underline{v}(r,\tau) + (1-\eta) \cdot \overline{v}(r,\tau)$

- i.e., Hurwicz (1951) preferences

Optimal Choices for a Given Group

Example 1: One Group

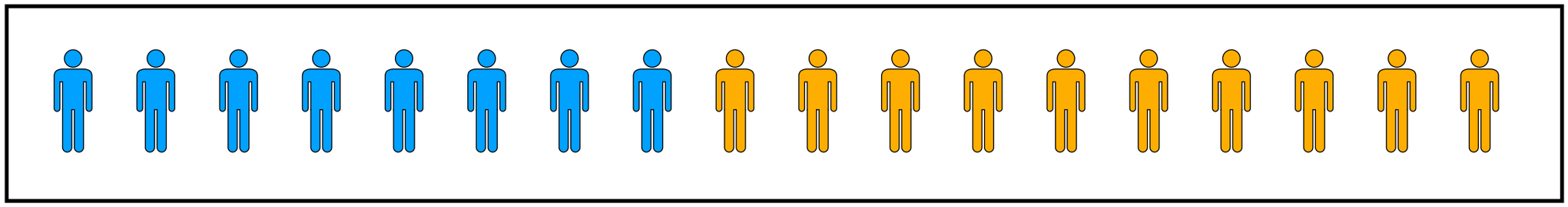


What the designer knows:

- 4/9 of patients benefit from treatment
- does not know **which** patients

What should a designer who assigns weight $\eta = 5/8$ to the worst case do?

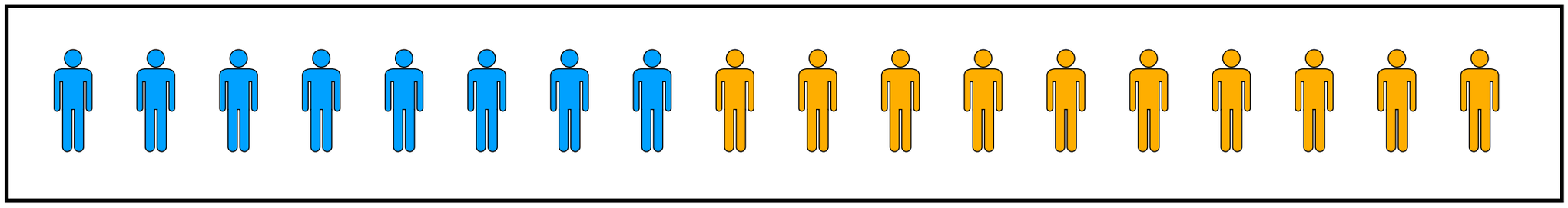
Example 1: One Group



4/9 of patients benefit from treatment

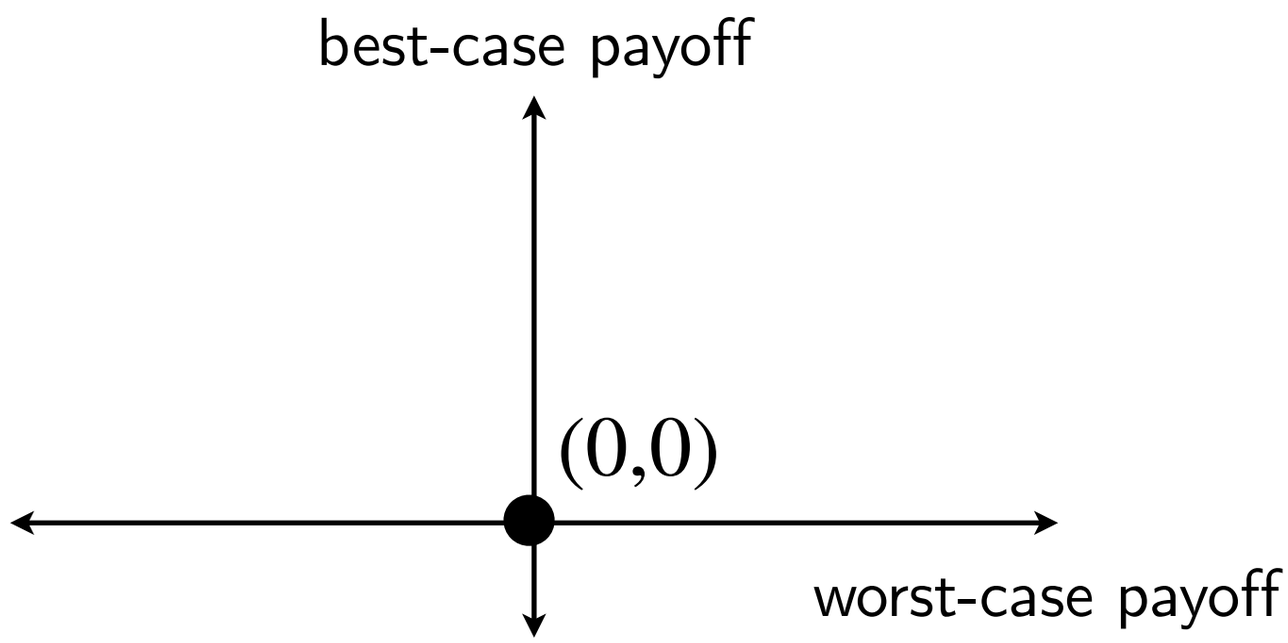
With no further information, treat **no one** (set $r = 0$) and receive a payoff of zero in both the worst and best case (**distrust point**).

Example 1: One Group

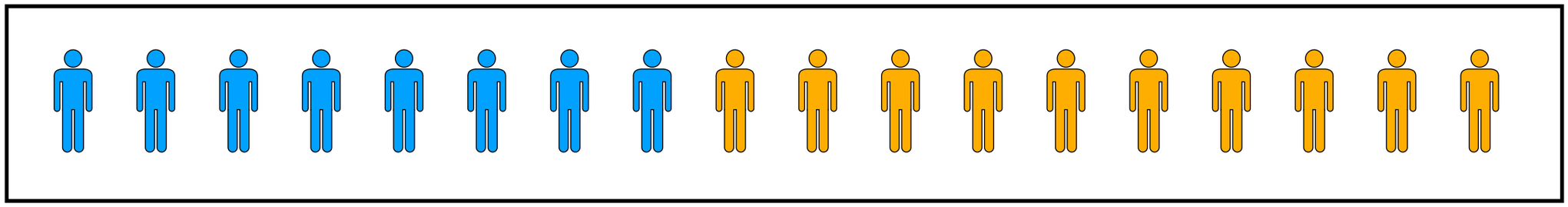


4/9 of patients benefit from treatment

With no further information, treat **no one** (set $r = 0$) and receive a payoff of zero in both the worst and best case (**distrust point**).



Example 1: One Group



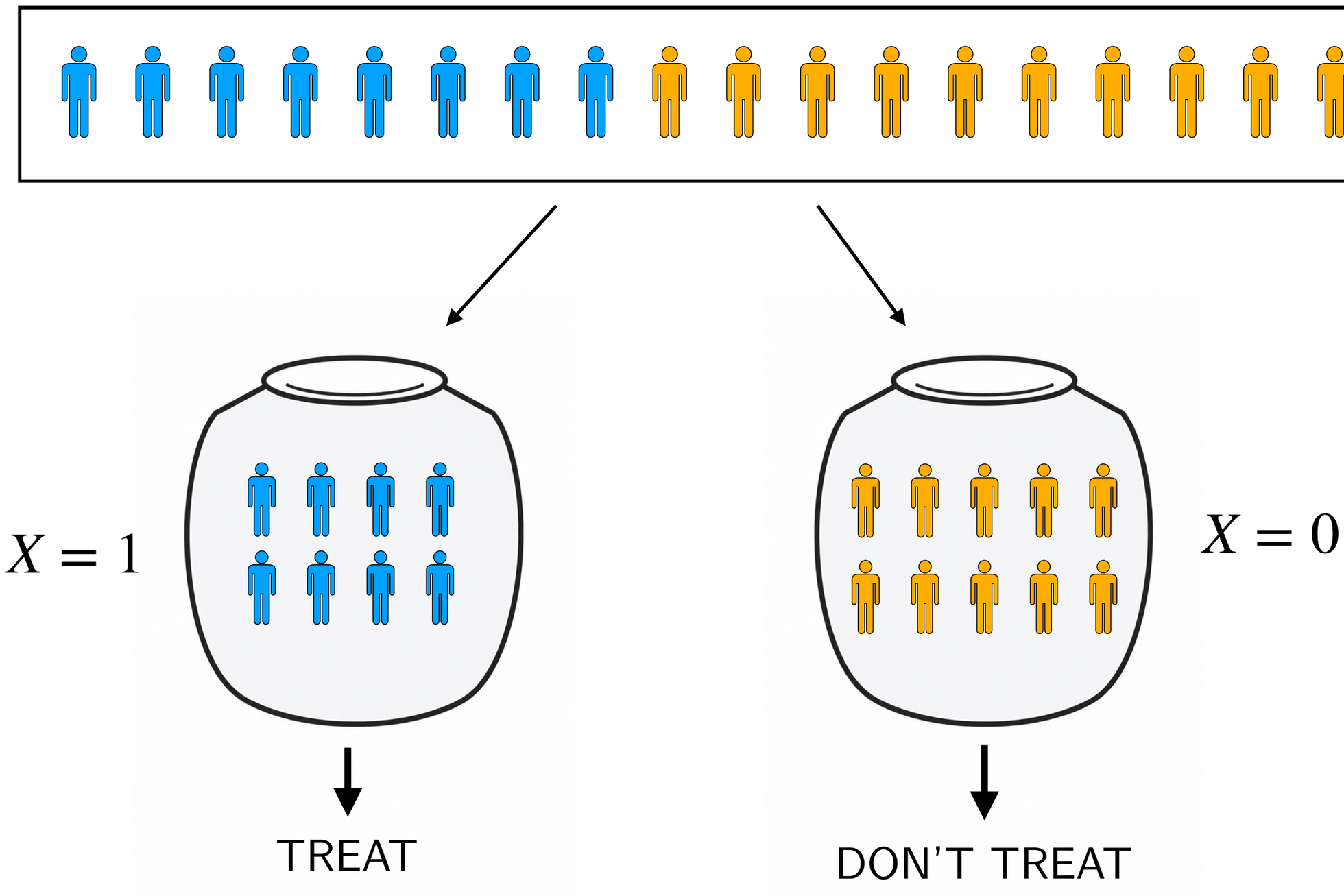
4/9 of patients benefit from treatment

With no further information, treat **no one** (set $r = 0$) and receive a payoff of zero in both the worst and best case (**distrust point**).

Consider instead:

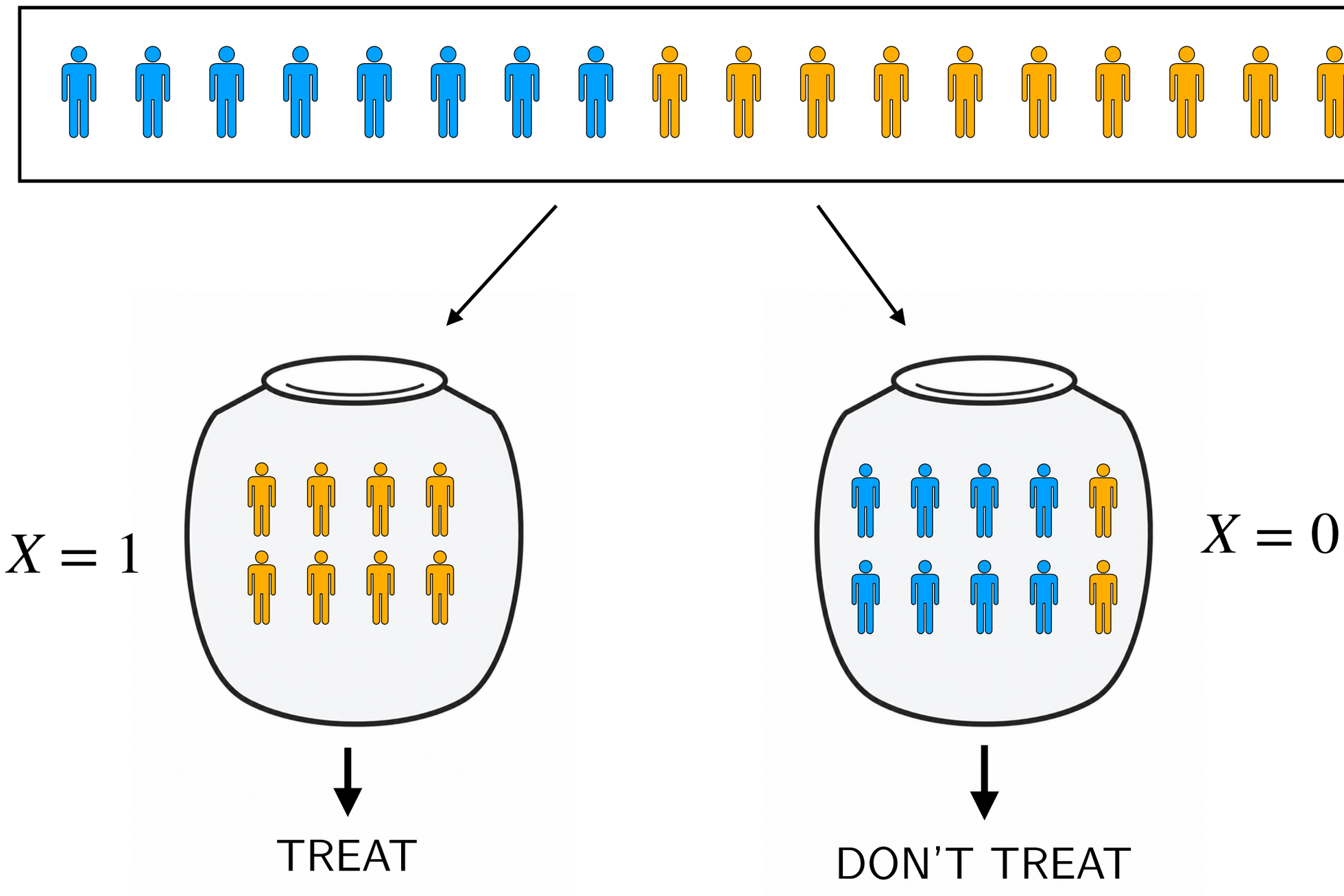
- give the AI access to a binary covariate $X \in \{0,1\}$ that is **unrestricted** in how informative it may be, i.e., $(\underline{\tau}, \overline{\tau}) = (0,1)$
- require treatment of $r = 4/9$ patients

Best Case



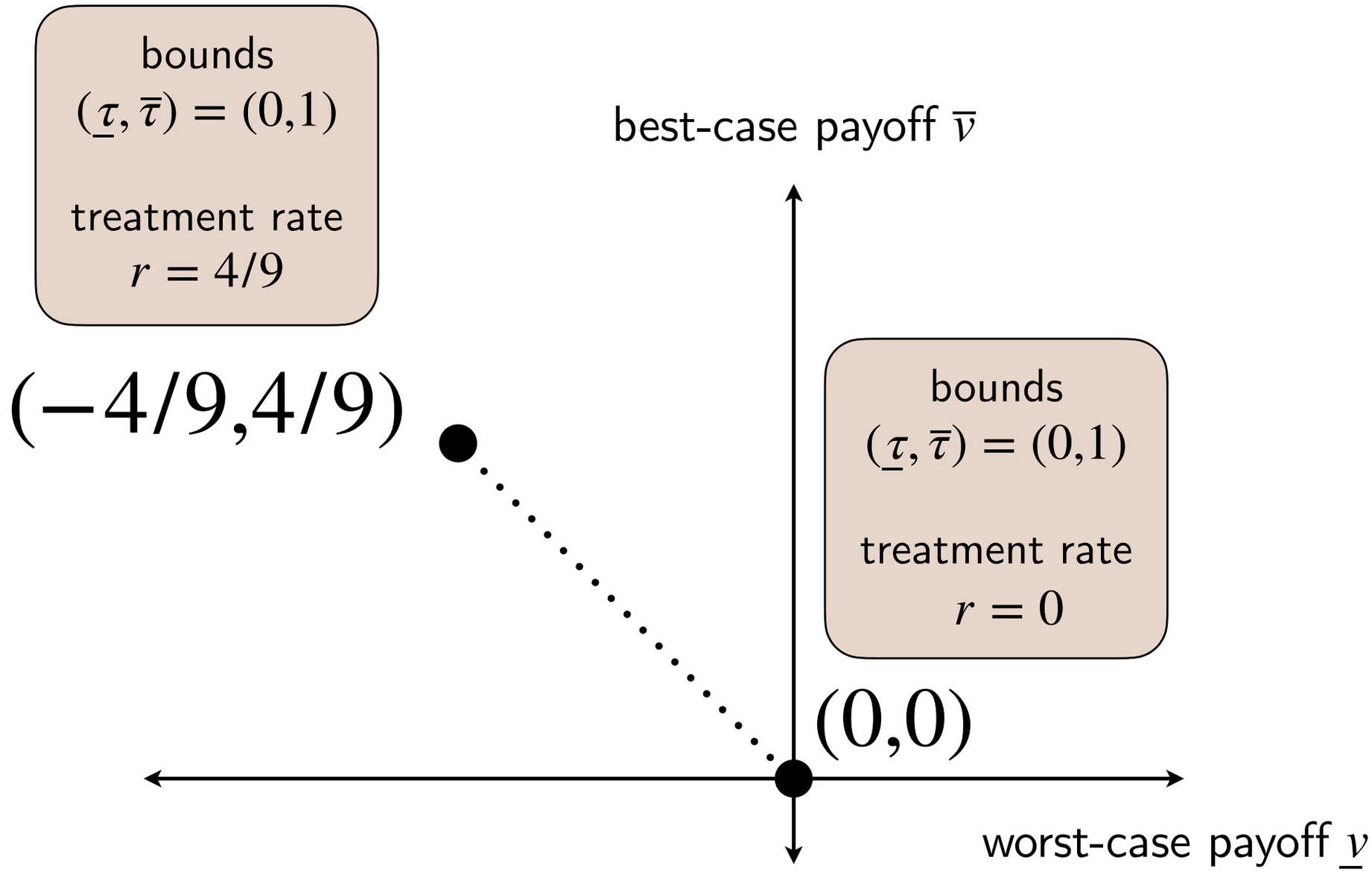
The AI precisely treats those patients who benefit: designer's payoff is **4/9**.

Worst Case

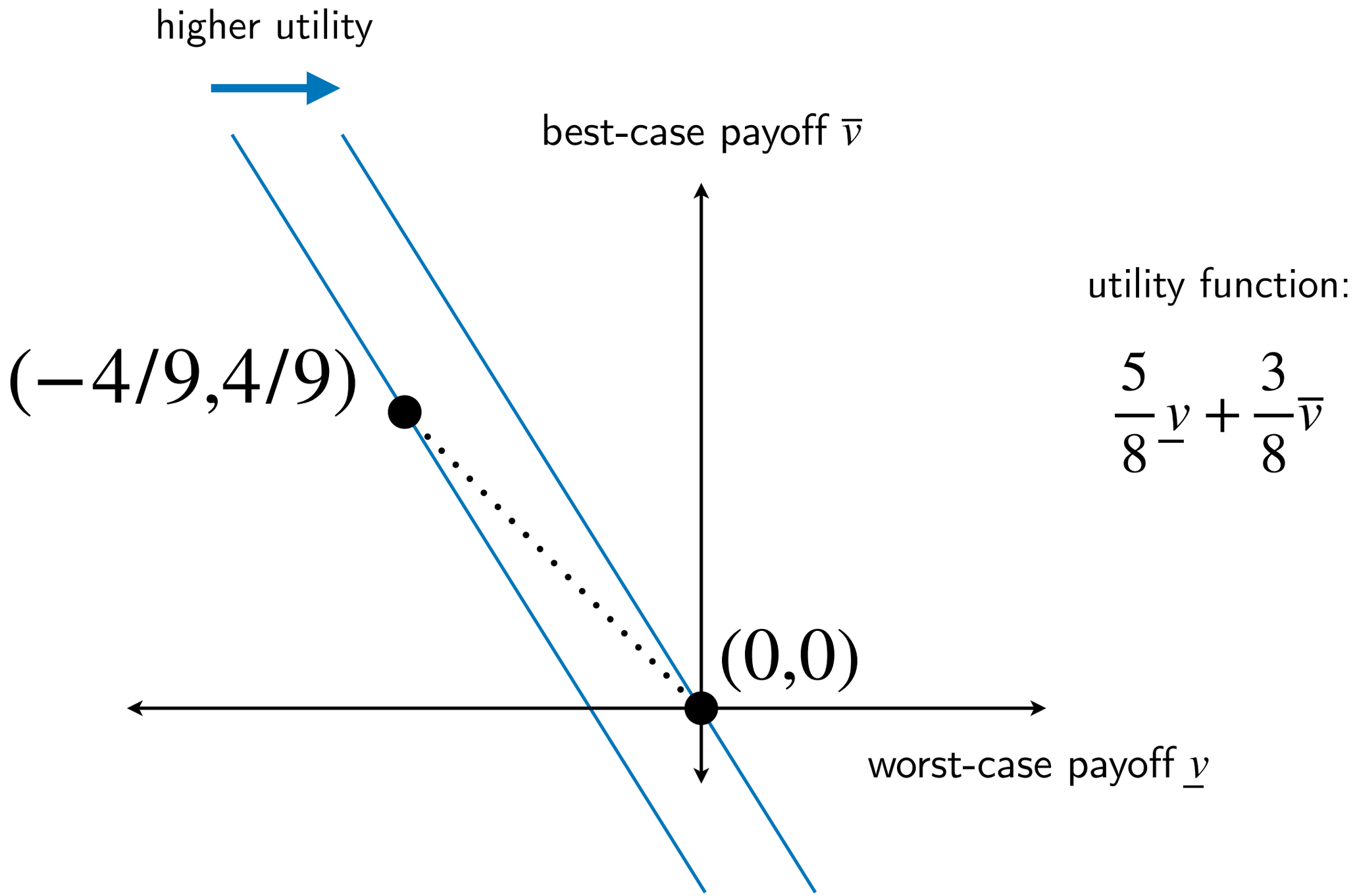


The AI treats only patients who do not benefit: designer's payoff is **-4/9**.

Best-Case and Worst-Case Payoffs



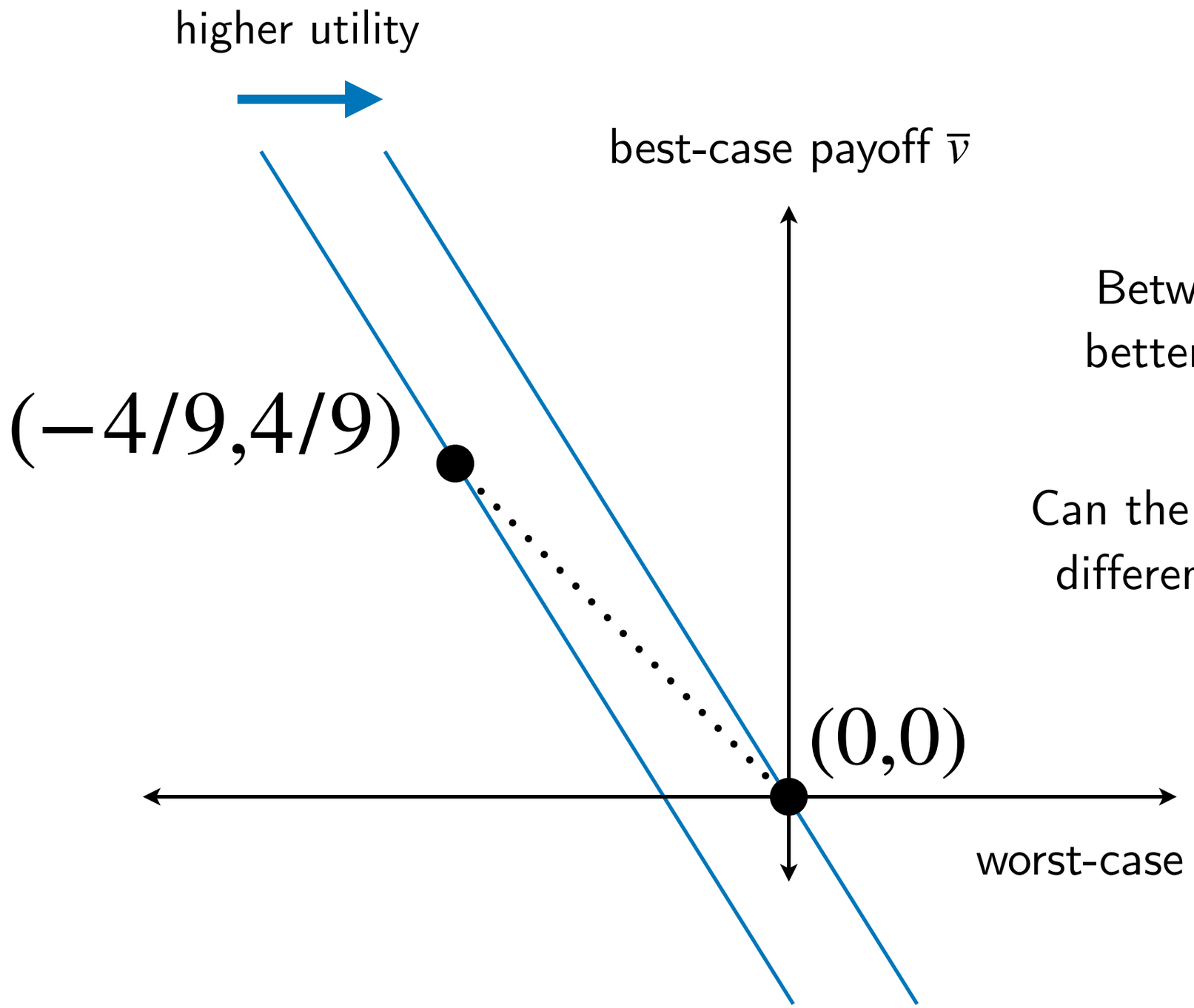
Best-Case and Worst-Case Payoffs



utility function:

$$\frac{5}{8}\underline{v} + \frac{3}{8}\overline{v}$$

Best-Case and Worst-Case Payoffs



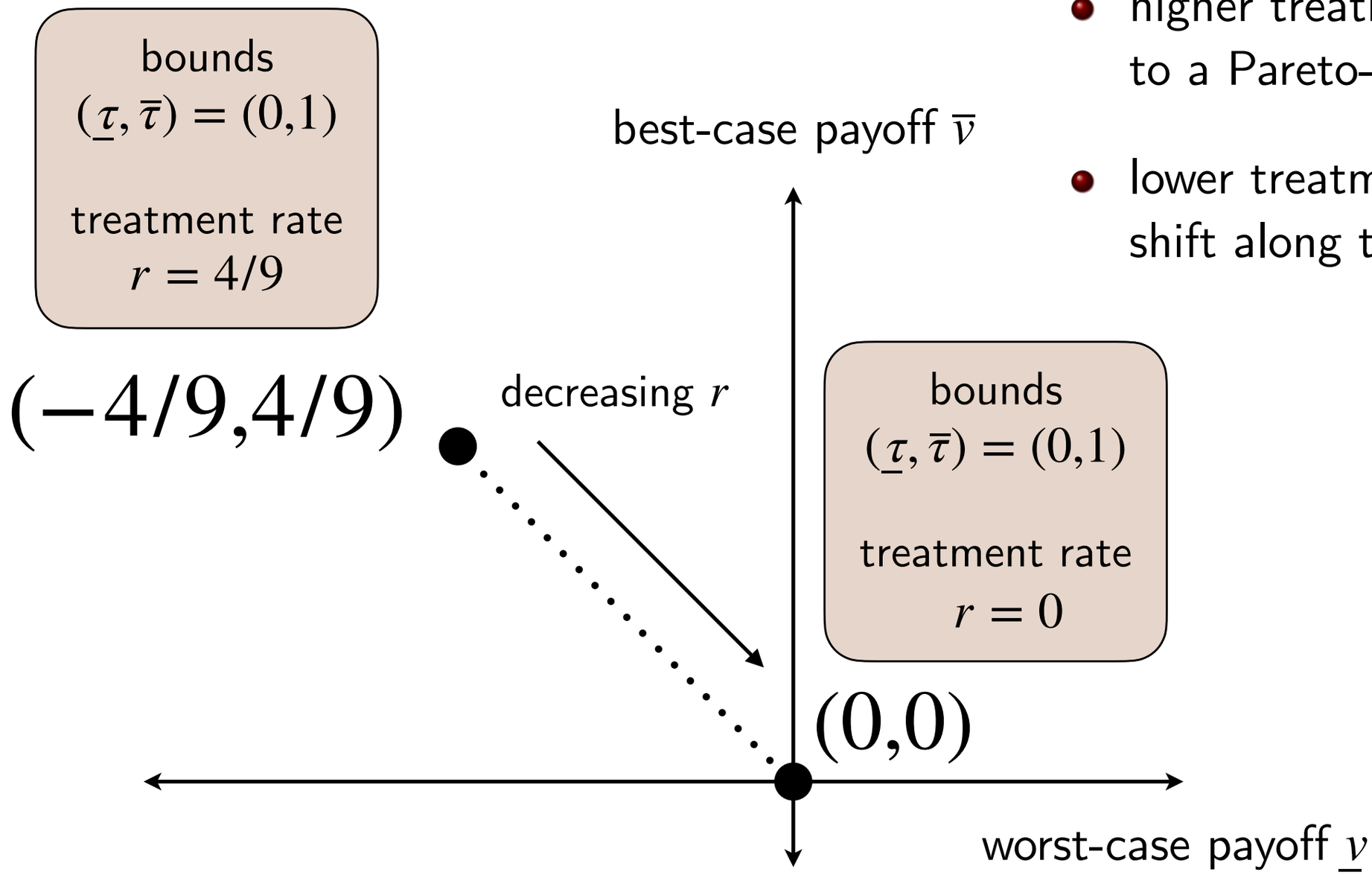
Between these two options, better not to delegate at all.

Can the designer do better with a different choice of $(\underline{\tau}, \overline{\tau})$ and r?

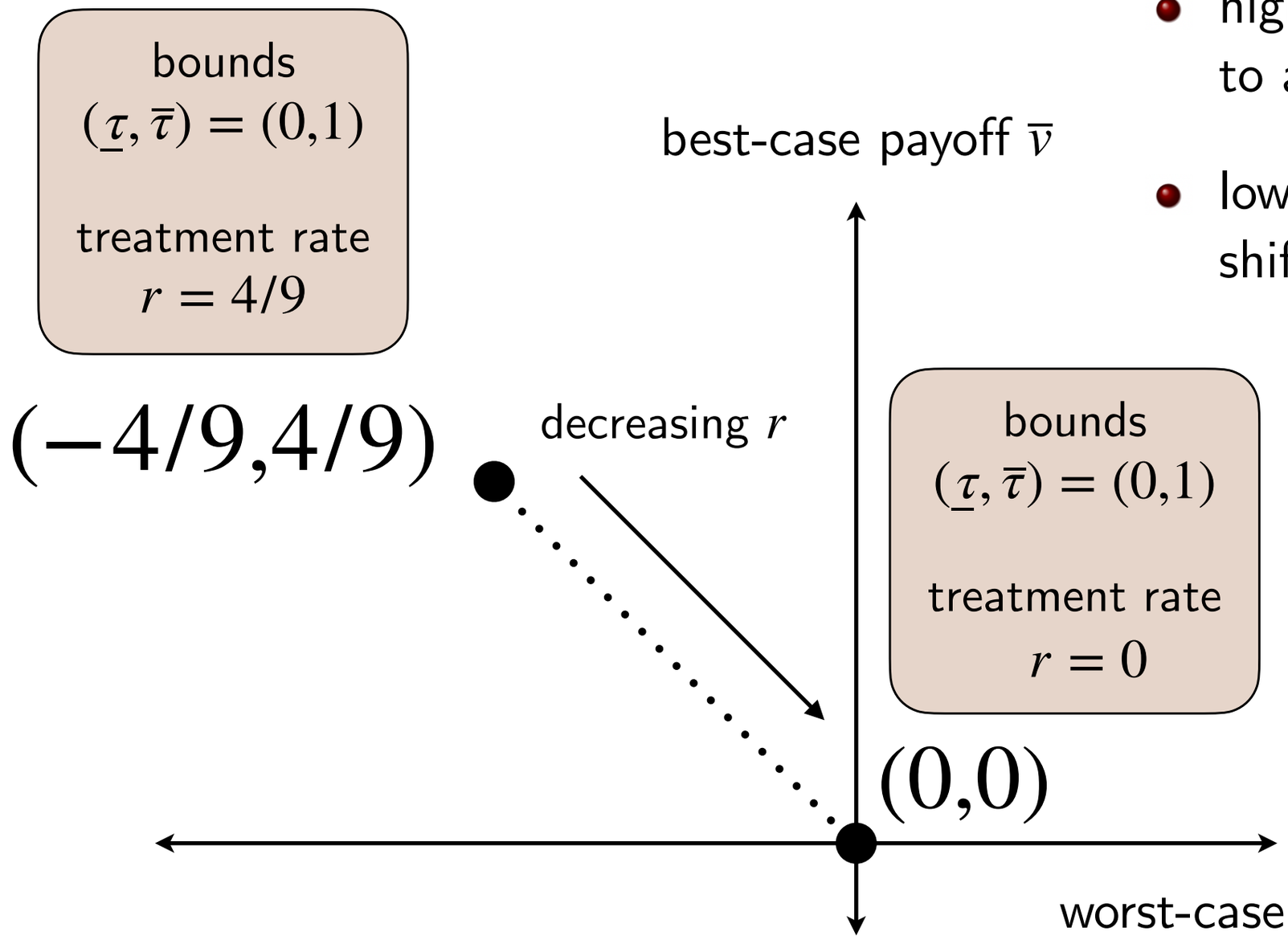
Best-Case and Worst-Case Payoffs

Not by changing r alone

- higher treatment rates lead to a Pareto-dominated point
- lower treatment rates just shift along this line.



Best-Case and Worst-Case Payoffs



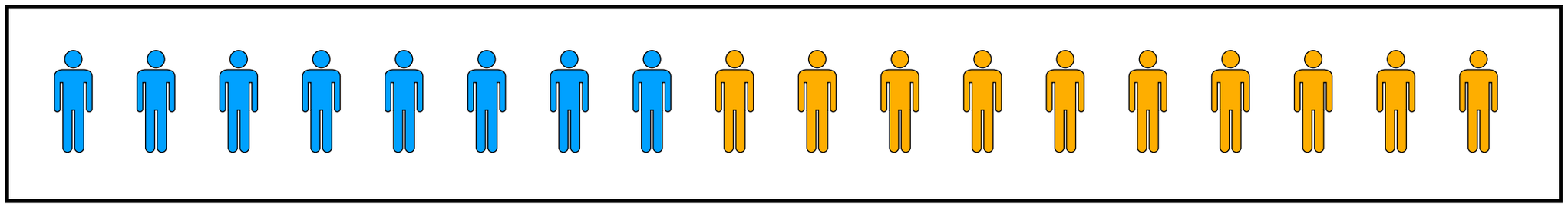
Not by changing r alone

- higher treatment rates lead to a Pareto-dominated point
- lower treatment rates just shift along this line.

But the designer can achieve a better tradeoff via choice of τ

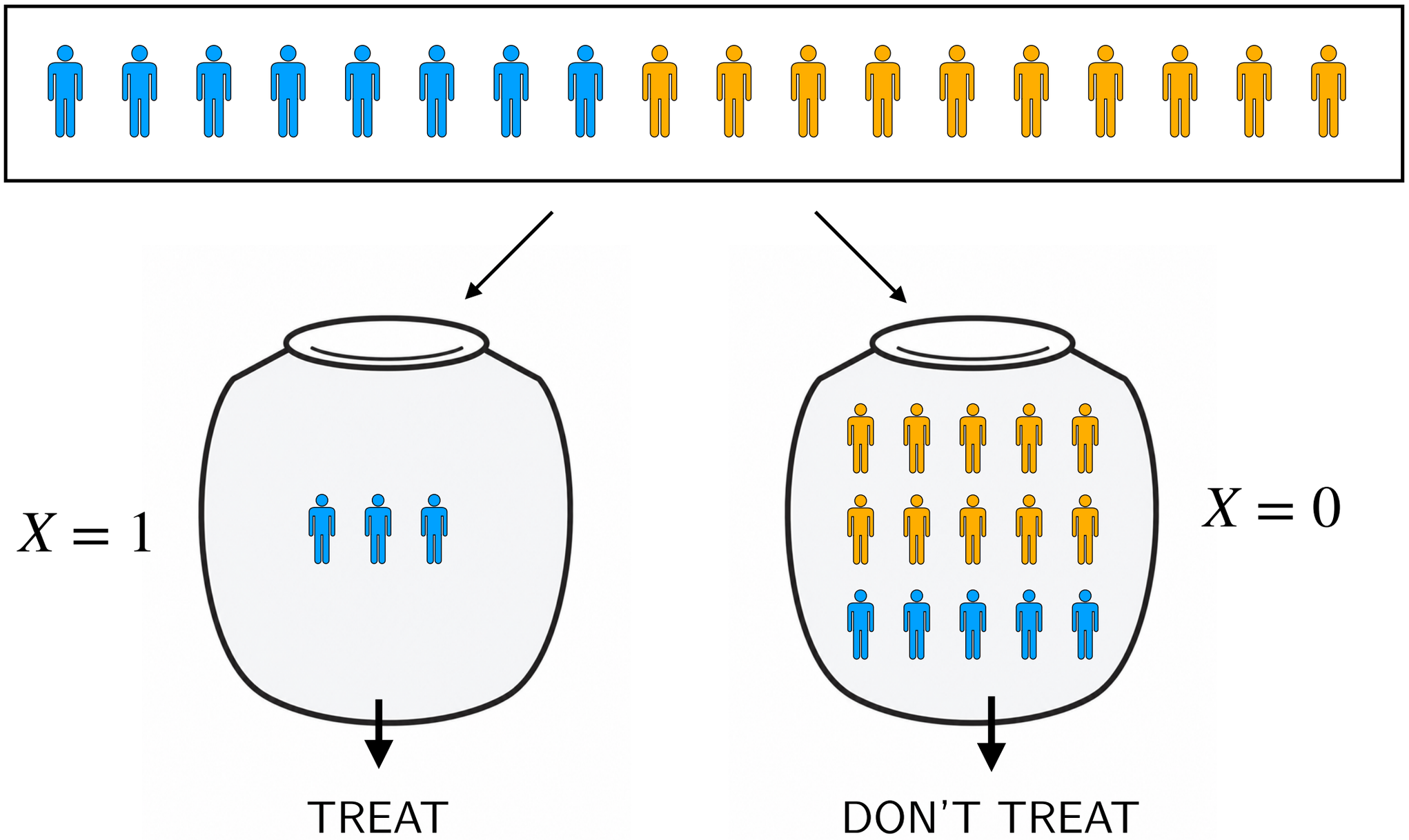
Example 1: One Group

Let $(\underline{\tau}, \overline{\tau}) = (1/3, 1)$ and $r = 3/18$.



Example 1: One Group

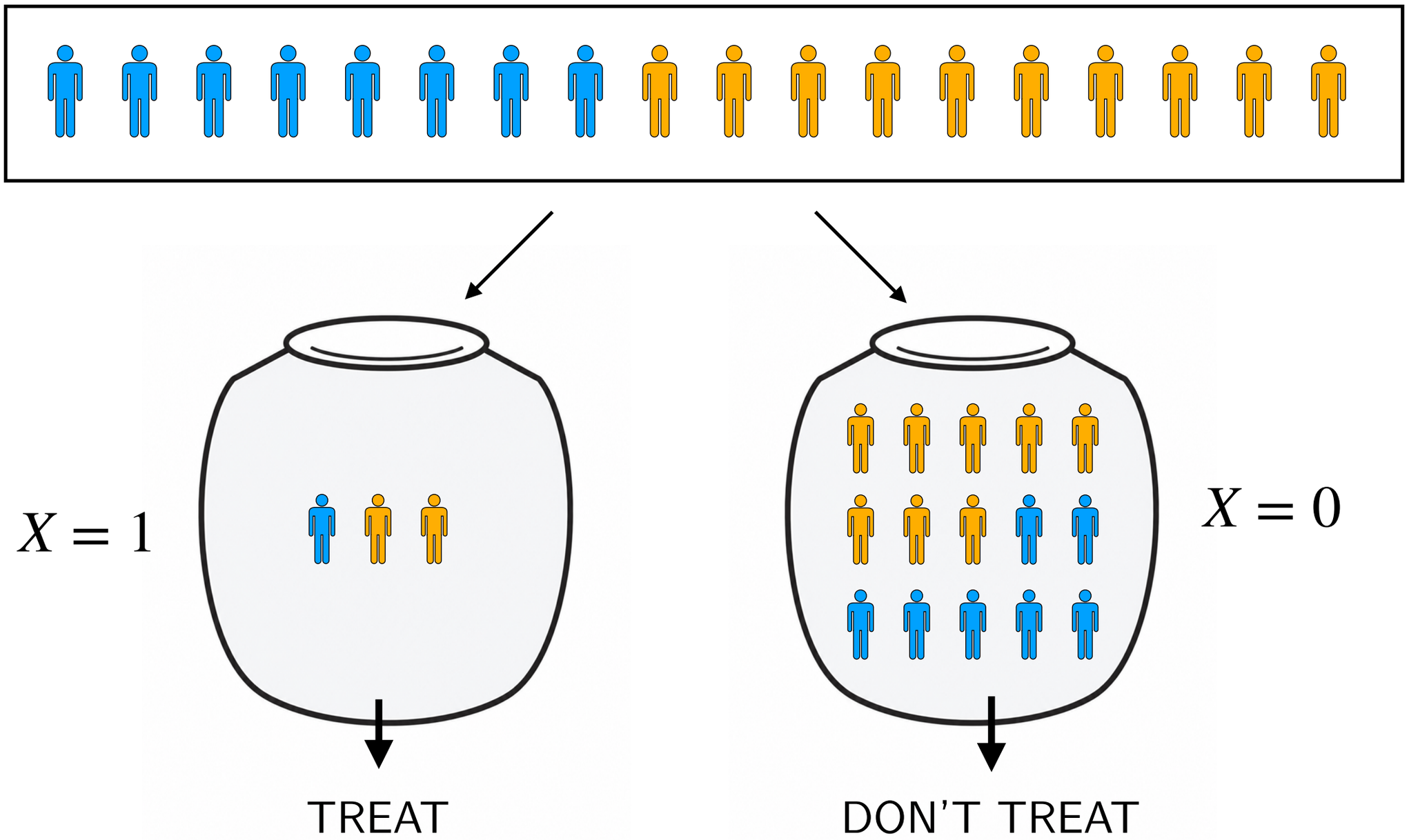
Let $(\underline{\tau}, \overline{\tau}) = (1/3, 1)$ and $r = 3/18$.



Treat 3/18 of patients, all of whom benefit: payoff = **3/18**

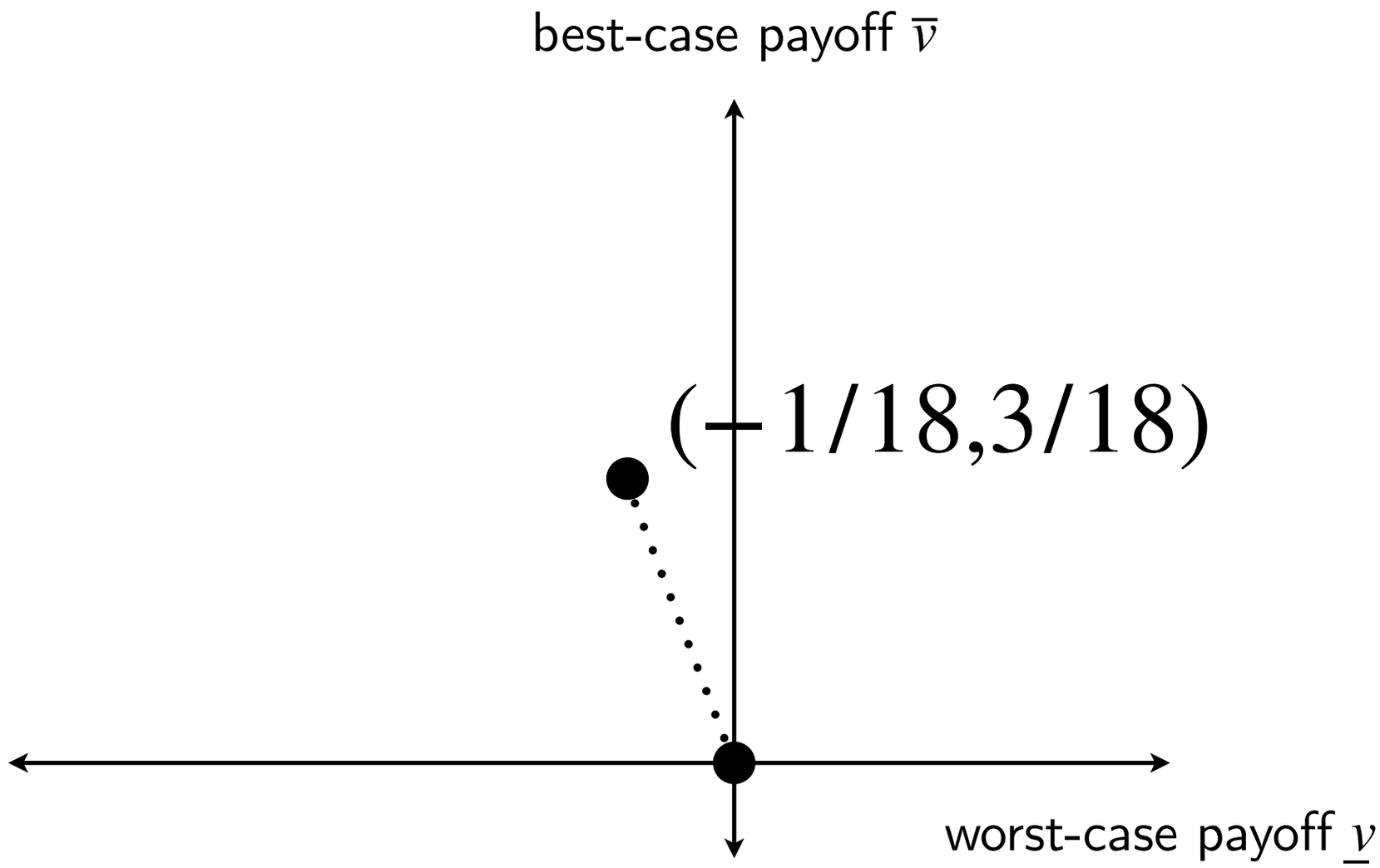
Example 1: One Group

Let $(\underline{\tau}, \overline{\tau}) = (1/3, 1)$ and $r = 3/18$.

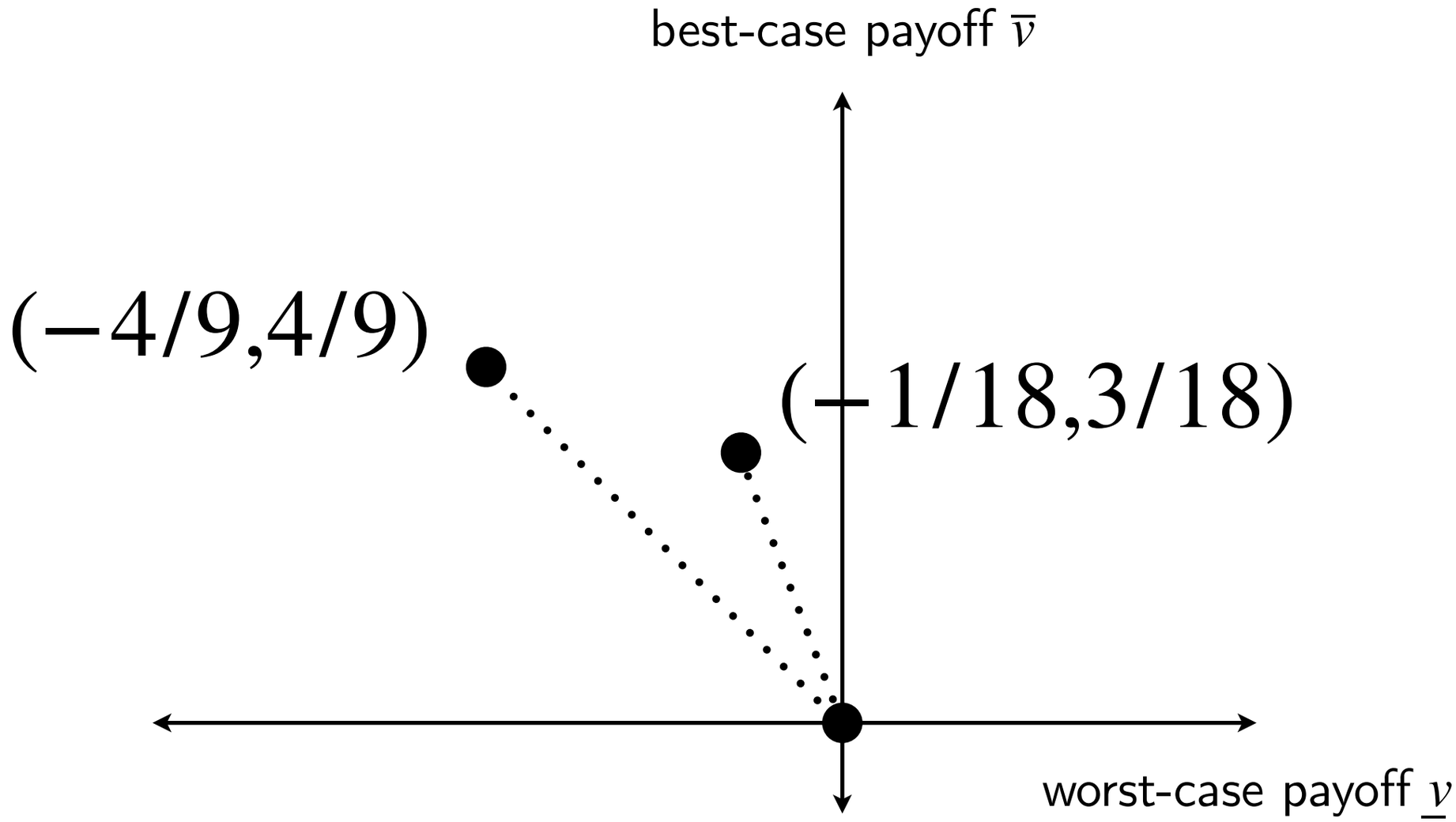


Treat 3/18 of patients, 1/3 of whom benefit: payoff = **-1/18**

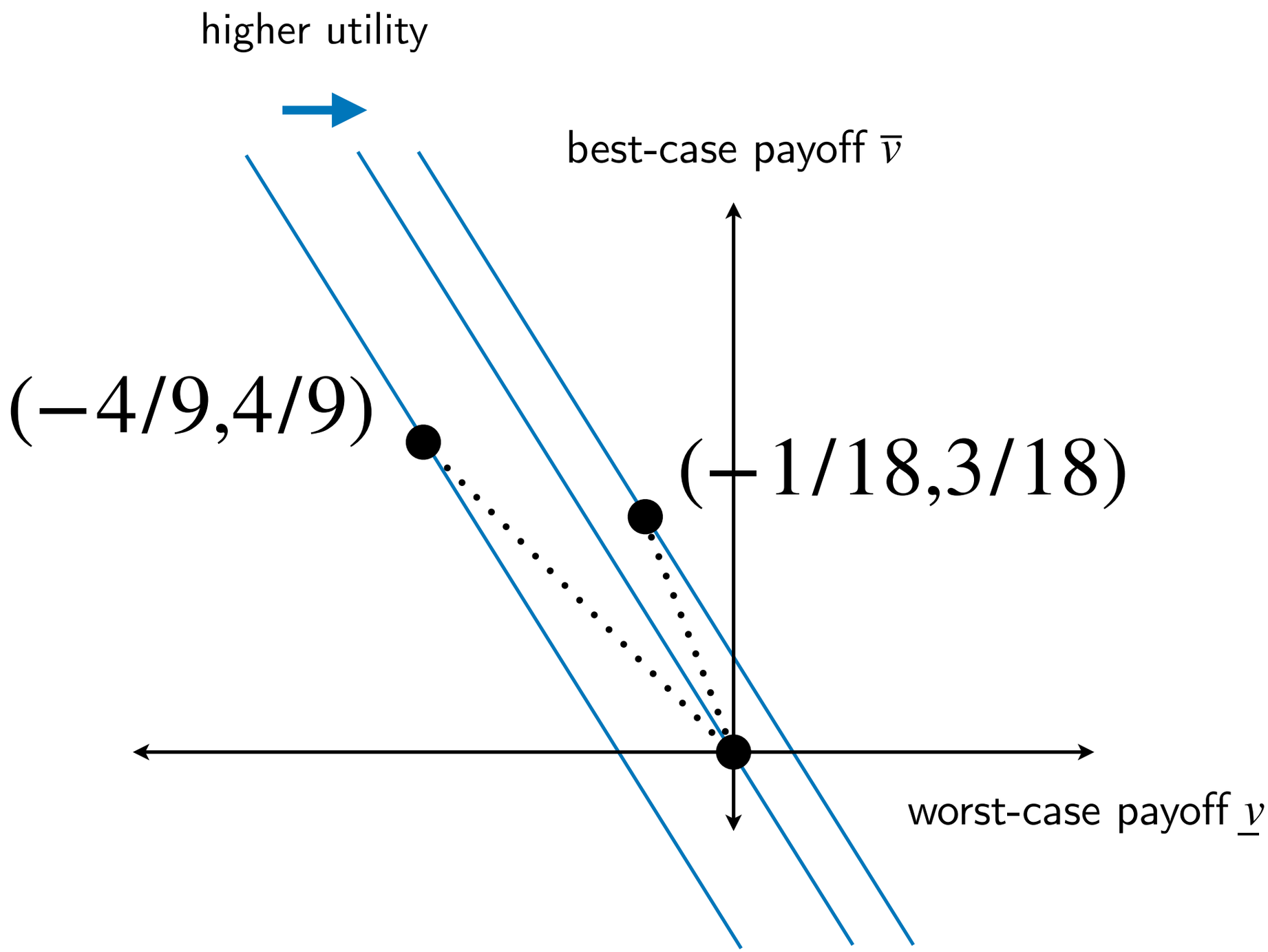
Best-Case and Worst-Case Payoffs



Best-Case and Worst-Case Payoffs



Best-Case and Worst-Case Payoffs



Optimal Design for Each Group

In each group g, the designer either optimally implements a **distrust point** or a **reliance point**.

DISTRUST POINT:

- Set the treatment rate to 1 if $p_g \geq 1/2$ and 0 otherwise
- Yields worst-case payoff = best case payoff = $\max\{0, 2p_g - 1\}$

Optimal Design for Each Group

In each group g, the designer either optimally implements a **distrust point** or a **reliance point**.

DISTRUST POINT:

- Set the treatment rate to 1 if $p_g \geq 1/2$ and 0 otherwise
- Yields worst-case payoff = best case payoff = $\max\{0, 2p_g - 1\}$

RELIANCE POINT: Choose (τ_g^*, r_g^*) where $r_g^* = (p_g - \underline{\tau}_g^*)/(\bar{\tau}_g^* - \underline{\tau}_g^*)$ and

- $\bar{\tau}_g^* = 1$ if $p_g < 1/2$ and $\underline{\tau}_g^* = 0$ if $p_g > 1/2$.
- The other endpoint is optimally set to its extreme when $\eta = 0$, and monotonically moves towards p_g as η grows larger.
- (See paper for exact expressions.)

Intuition for $p_g < 1/2$

- The aligned AI wants to identify patients who benefit from treatment. **Allow it** (set $\overline{\tau}_g^* = 1$).
- The misaligned AI wants to identify patients who are harmed by treatment. **Constrain this** (set $\underline{\tau}_g^* > 0$).

Intuition for $p_g < 1/2$

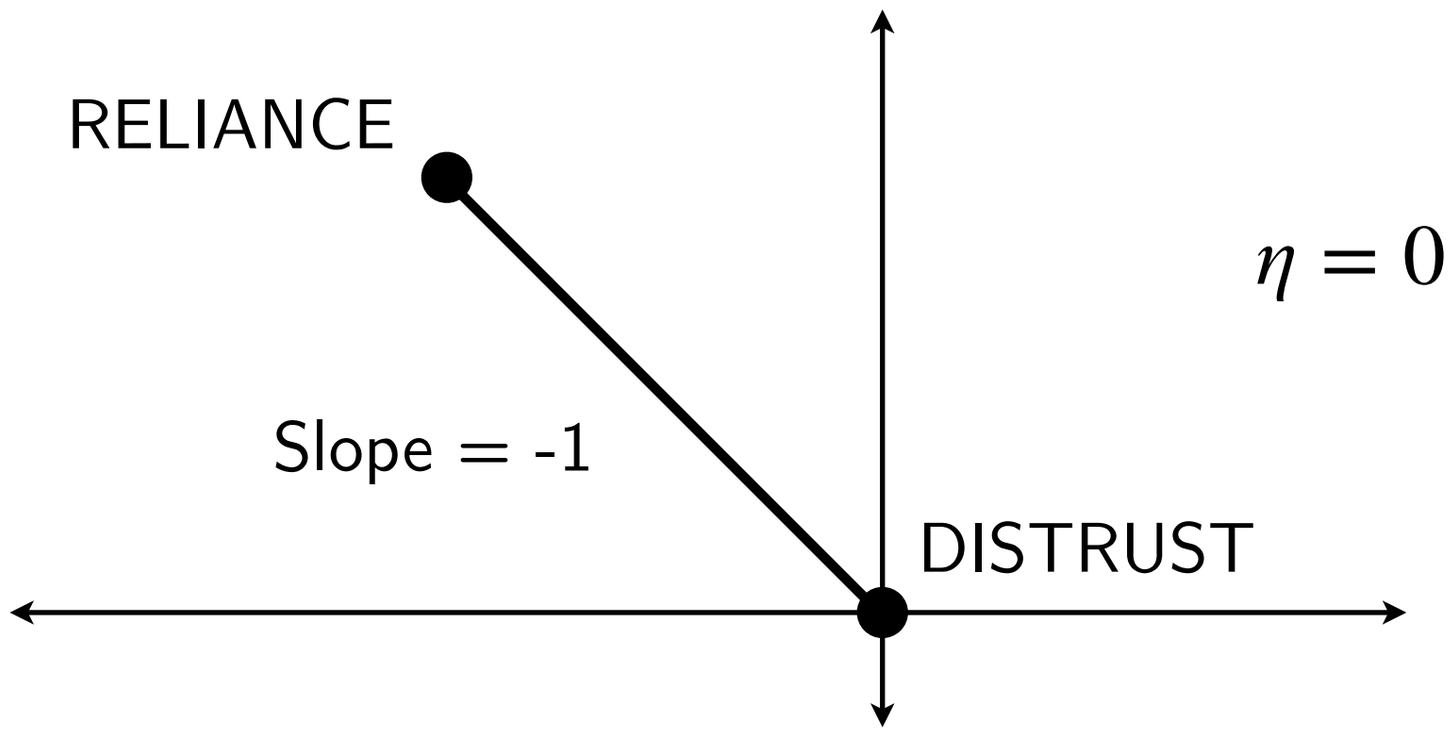
- The aligned AI wants to identify patients who benefit from treatment. **Allow it** (set $\overline{\tau}_g^* = 1$).
- The misaligned AI wants to identify patients who are harmed by treatment. **Constrain this** (set $\underline{\tau}_g^* > 0$).
- But the latter has a cost—limits how many patients can be correctly identified who benefit from treatment
- To see this, consider a binary X and suppose

$$P(Y = 1 \mid X = 1) = \overline{\tau}_g^* \qquad P(Y = 1 \mid X = 0) = \underline{\tau}_g^*$$

- Then $P(X = 1)\overline{\tau}_g^* + P(X = 0)\underline{\tau}_g^* = p_g$ by the law of total probability. Increasing $\underline{\tau}_g^*$ forces $P(X = 1)$ down $\rightarrow$ reduces the best case
- The optimal treatment rate r^* is the largest mass of patients that can be assigned to the higher conditional probability $\overline{\tau}_g^*$

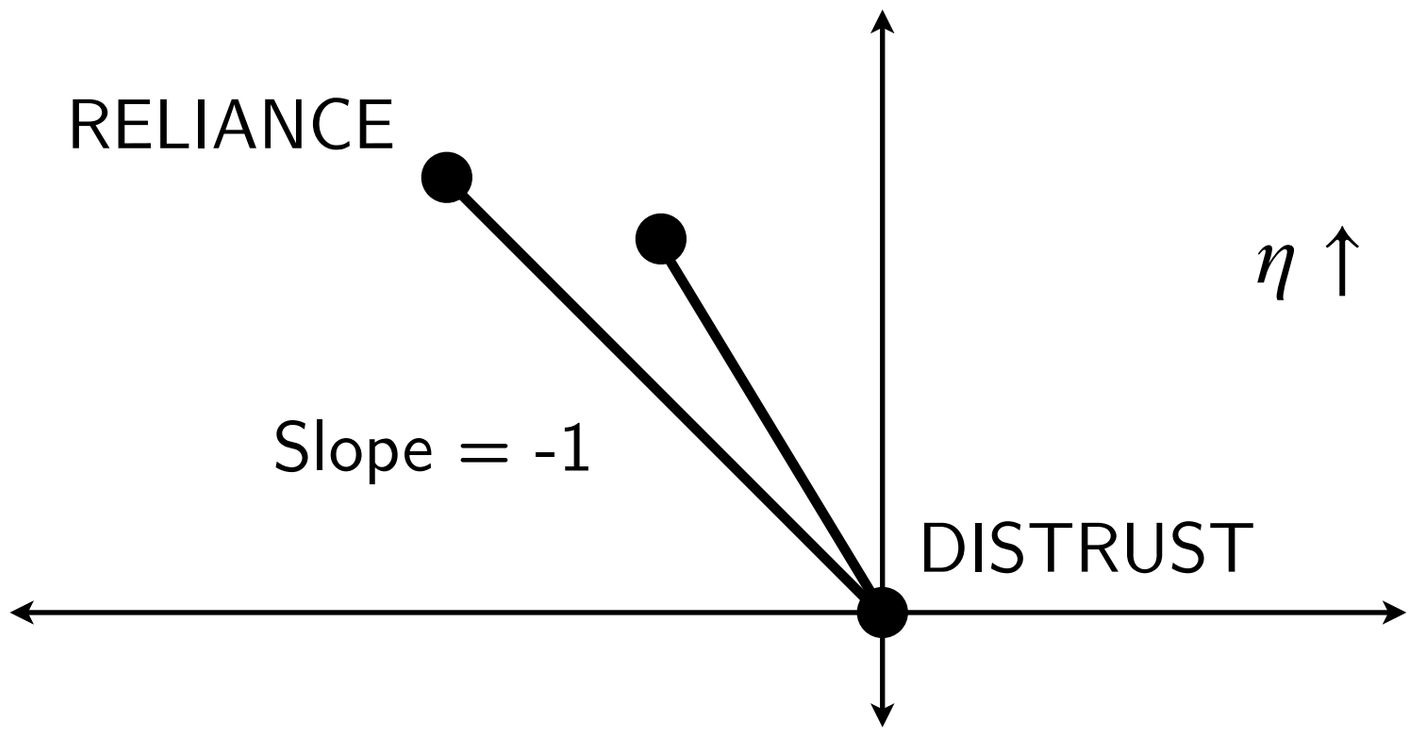
Intuition for $p_g < 1/2$

- Choice of $\underline{\tau}_g^*$ needs to tradeoff between
  - improving the best case/worst case **tradeoff** (larger $\underline{\tau}_g^*$)
  - improving the **best case payoff** (smaller $\underline{\tau}_g^*$)
- The designer resolves this tradeoff given their weight on the worst case η



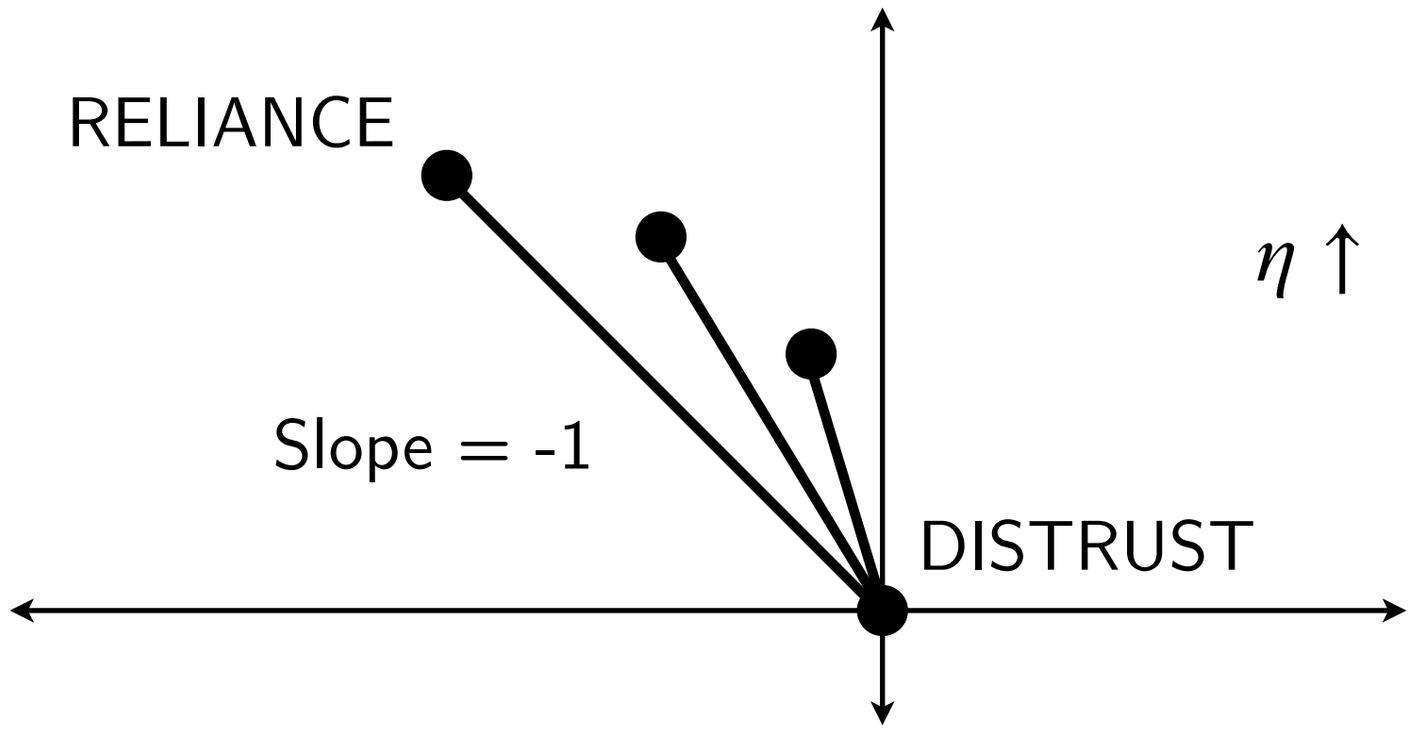
Intuition for $p_g < 1/2$

- Choice of $\underline{\tau}_g^*$ needs to tradeoff between
  - improving the best case/worst case **tradeoff** (larger $\underline{\tau}_g^*$)
  - improving the **best case payoff** (smaller $\underline{\tau}_g^*$)
- The designer resolves this tradeoff given their weight on the worst case η



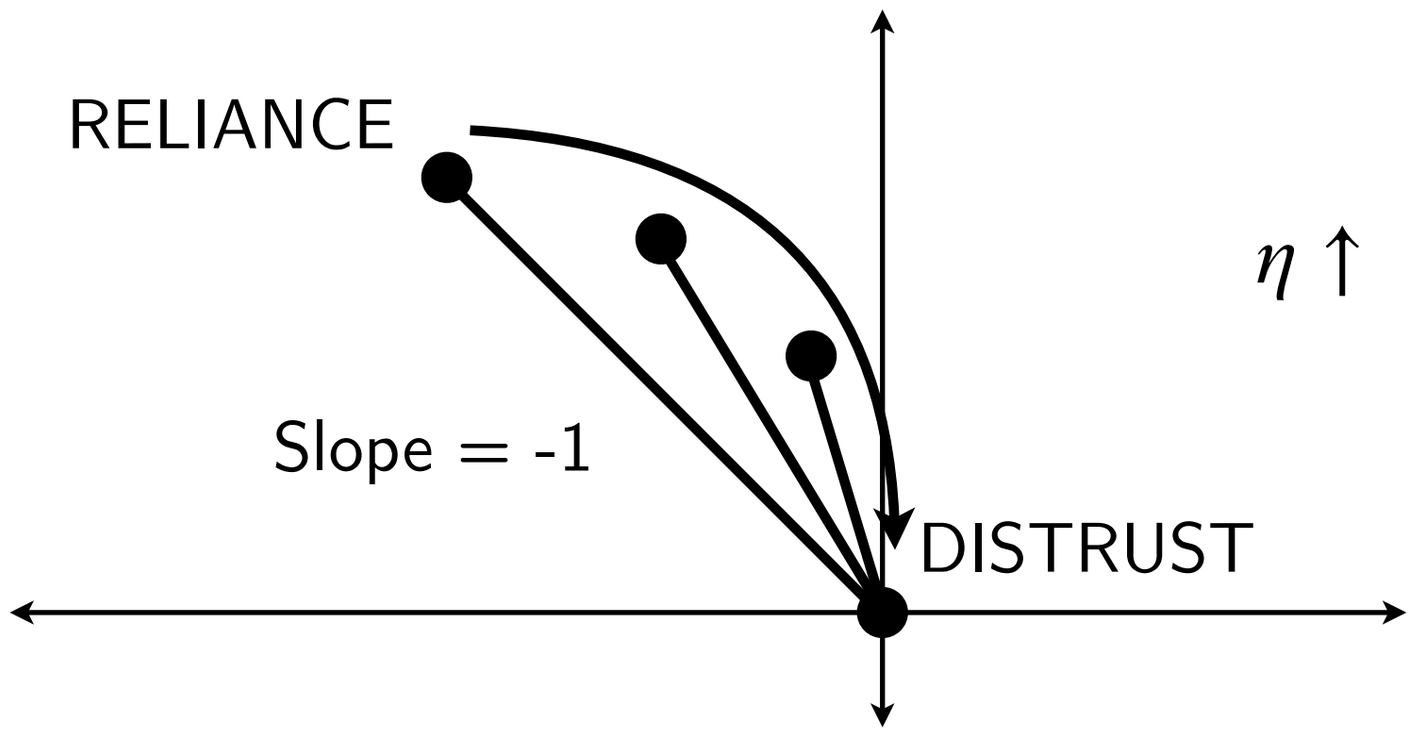
Intuition for $p_g < 1/2$

- Choice of $\underline{\tau}_g^*$ needs to tradeoff between
  - improving the best case/worst case **tradeoff** (larger $\underline{\tau}_g^*$)
  - improving the **best case payoff** (smaller $\underline{\tau}_g^*$)
- The designer resolves this tradeoff given their weight on the worst case η



Intuition for $p_g < 1/2$

- Choice of $\underline{\tau}_g^*$ needs to tradeoff between
  - improving the best case/worst case **tradeoff** (larger $\underline{\tau}_g^*$)
  - improving the **best case payoff** (smaller $\underline{\tau}_g^*$)
- The designer resolves this tradeoff given their weight on the worst case η



The Extensive Margin of Reliance Across Groups

Many groups

- Now allow for many groups, recalling that they are differentiated by

$$p_g = P(Y = 1 \mid G = g)$$

- For each group, the optimal designer choices are as described before, but how does reliance on the AI differ across these groups?

Group	Probability of treatment success
18-30	40%
31-65	20%
66+	10%

Many groups

Say that the designer **strictly relies** on the AI if the optimal point is not the distrust point.

Many groups

Say that the designer **strictly relies** on the AI if the optimal point is not the distrust point.

Proposition: Order the groups as $g^{(1)}, \ldots, g^{(n)}$ where

$$\left| p_{g^{(1)}} - \frac{1}{2} \right| \leq \left| p_{g^{(2)}} - \frac{1}{2} \right| \leq \cdots \leq \left| p_{g^{(n)}} - \frac{1}{2} \right|$$

breaking ties arbitrarily.

Then there is a threshold $T(\eta)$ such that the designer with weight η on the worst case strictly relies on the AI for exactly the first $T(\eta)$ groups.

Moreover, the threshold $T(\eta)$ is monotonically decreasing in η.

Intuition

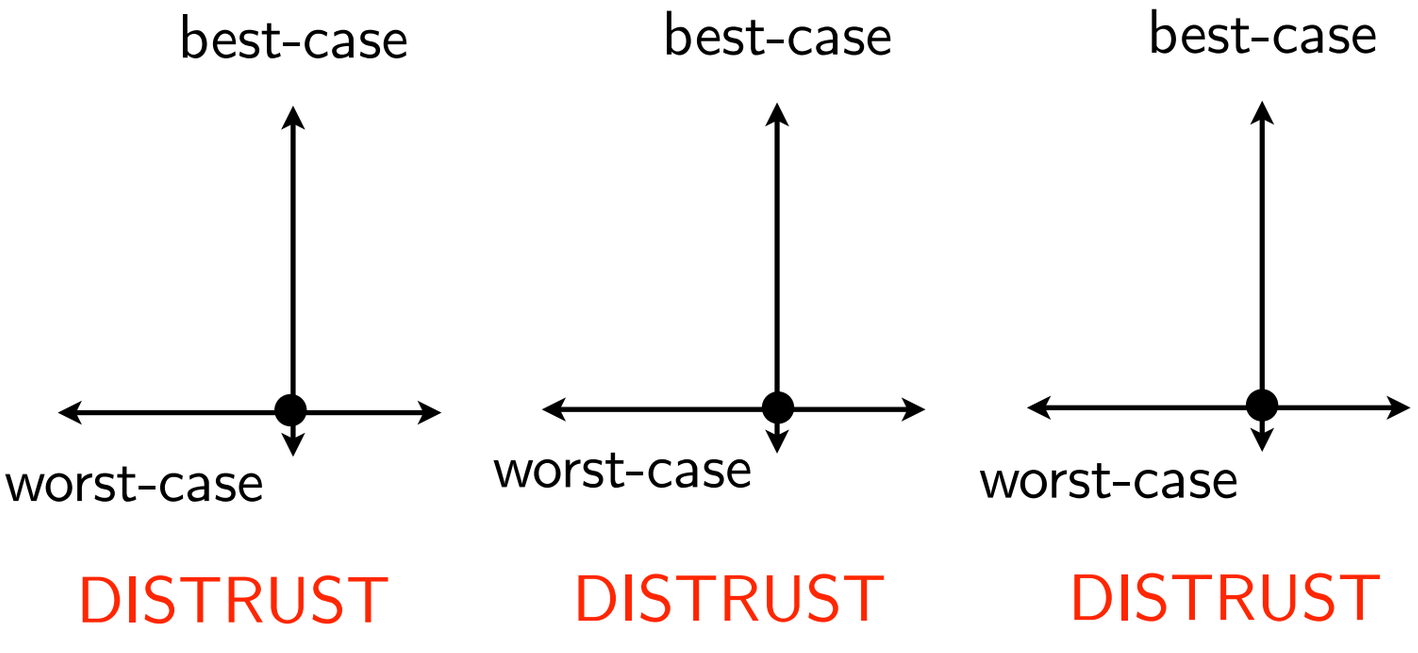
- Consider $p_g < 1/2$
- The main risk from delegating is that a misaligned AI concentrates treatment on individuals with low conditional success rates
- The designer limits this by increasing $\underline{\tau}_g$: regardless of how the AI allocate treatment, the probably that it is successful is at least $\underline{\tau}_g$
- But feasibility requires $\underline{\tau}_g \leq p_g$
- The closer the base rate is to $1/2$, the better the designer can protect against a misaligned AI
- For $p_g > 1/2$ the logic is symmetric but reversed

Example 2: Multiple Groups

	18-30	**31-65**	**66+**
p_g	0.4	0.2	0.1
$\left\lvert p_g - \frac{1}{2} \right\rvert$	0.1	0.3	0.4

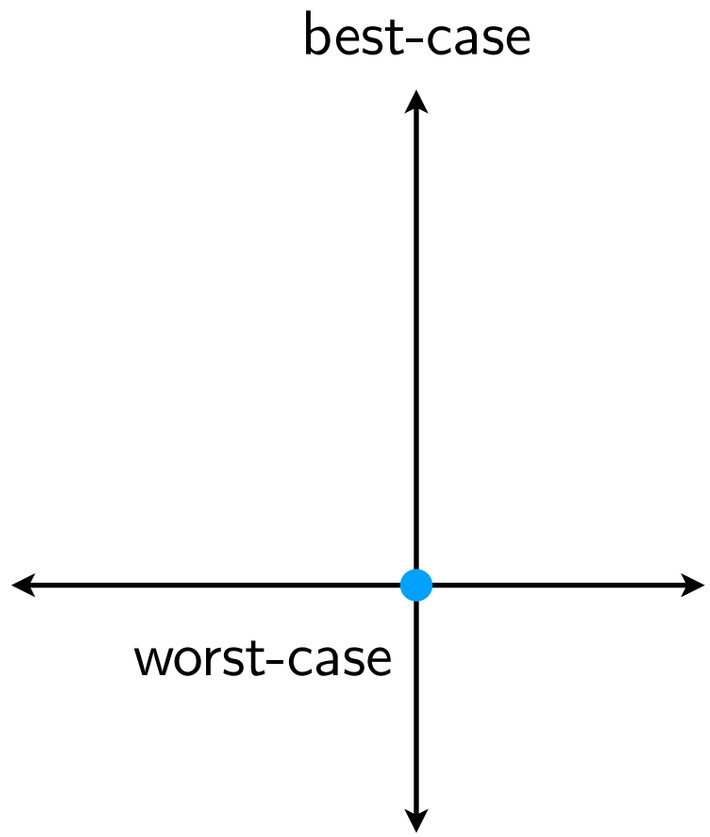
Example 2: Multiple Groups

	18-30	**31-65**	**66+**
p_g	0.4	0.2	0.1
$\left\lvert p_g - \frac{1}{2} \right\rvert$	0.1	0.3	0.4



$$\eta = 1$$

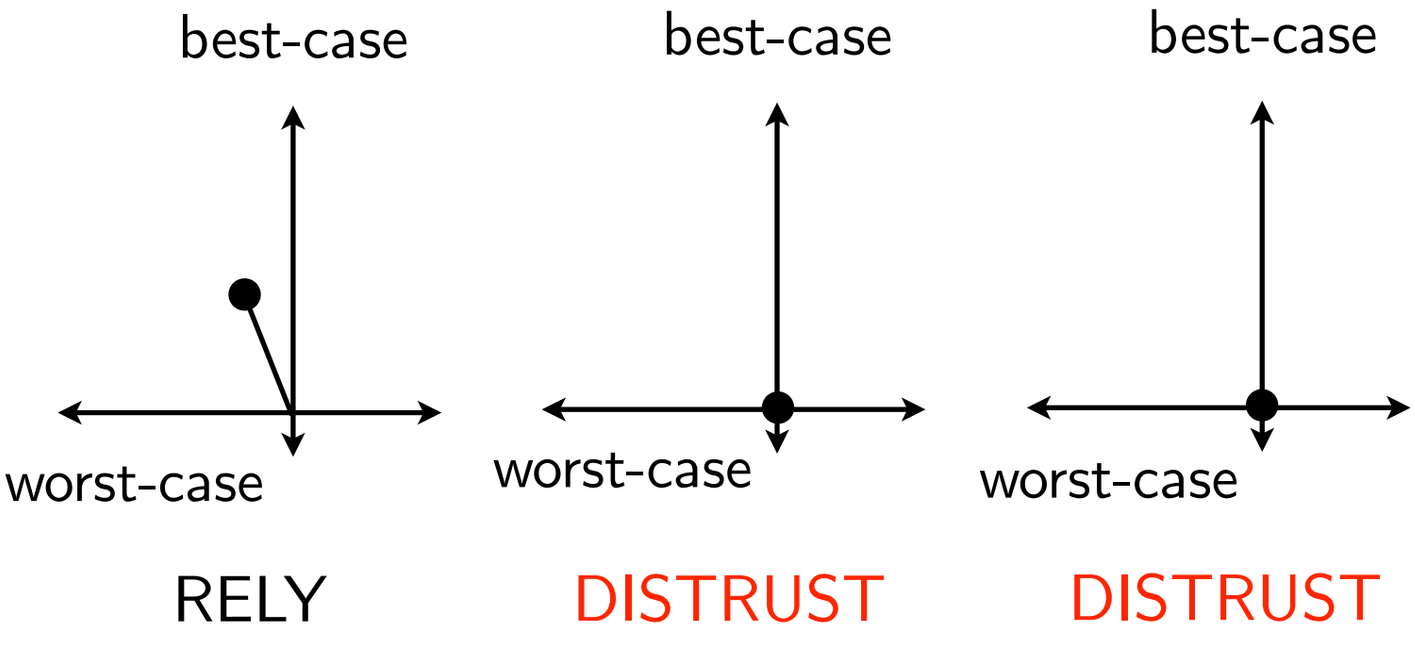
Aggregate Payoffs



(For the figure assume equal population sizes)

Example 2: Multiple Groups

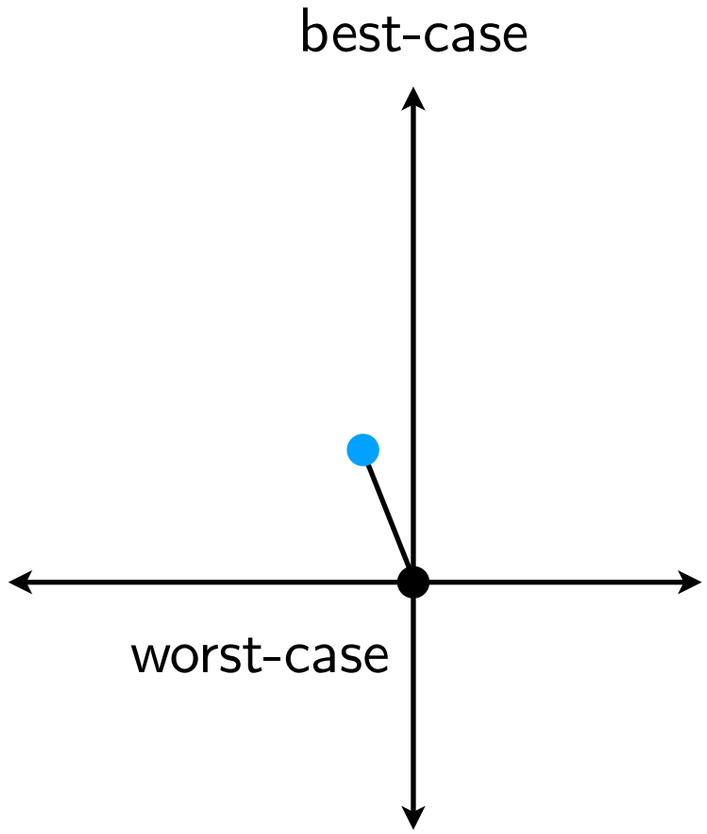
	18-30	**31-65**	**66+**
p_g	0.4	0.2	0.1
$\left\lvert p_g - \frac{1}{2} \right\rvert$	0.1	0.3	0.4



RELY DISTRUST DISTRUST

$\eta \downarrow$

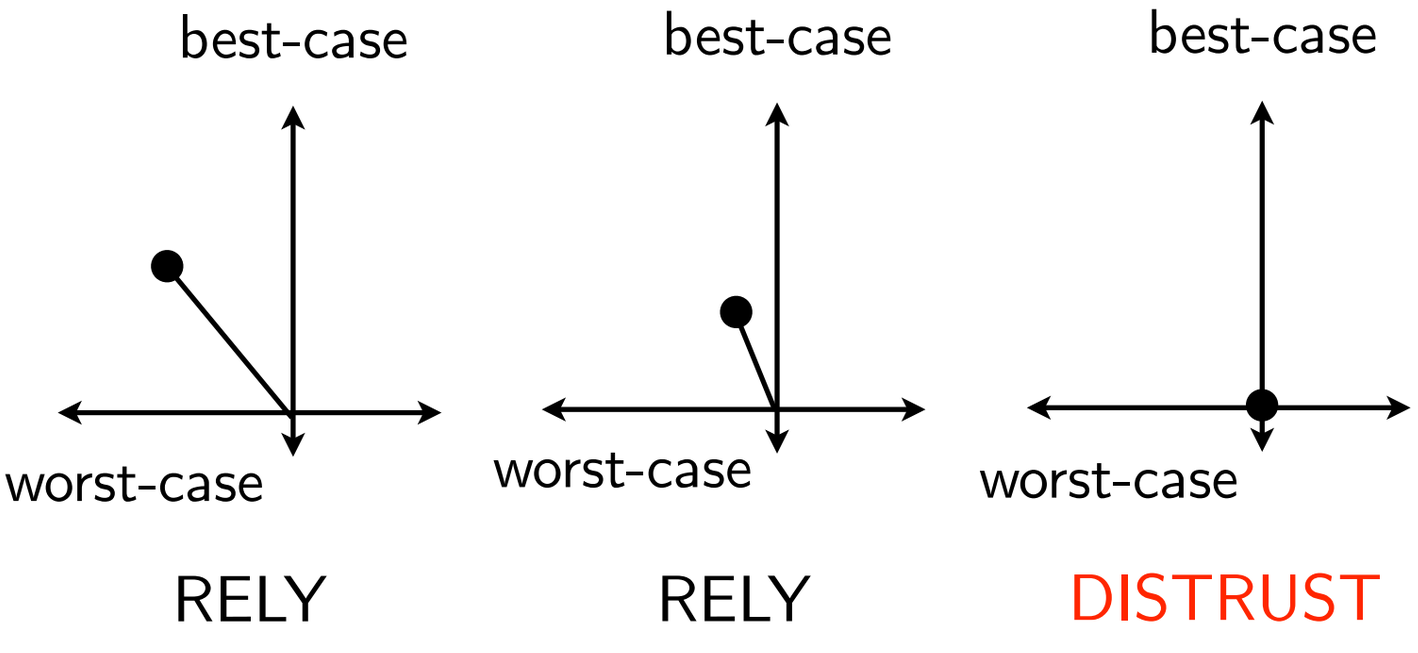
Aggregate Payoffs



(For the figure assume equal population sizes)

Example 2: Multiple Groups

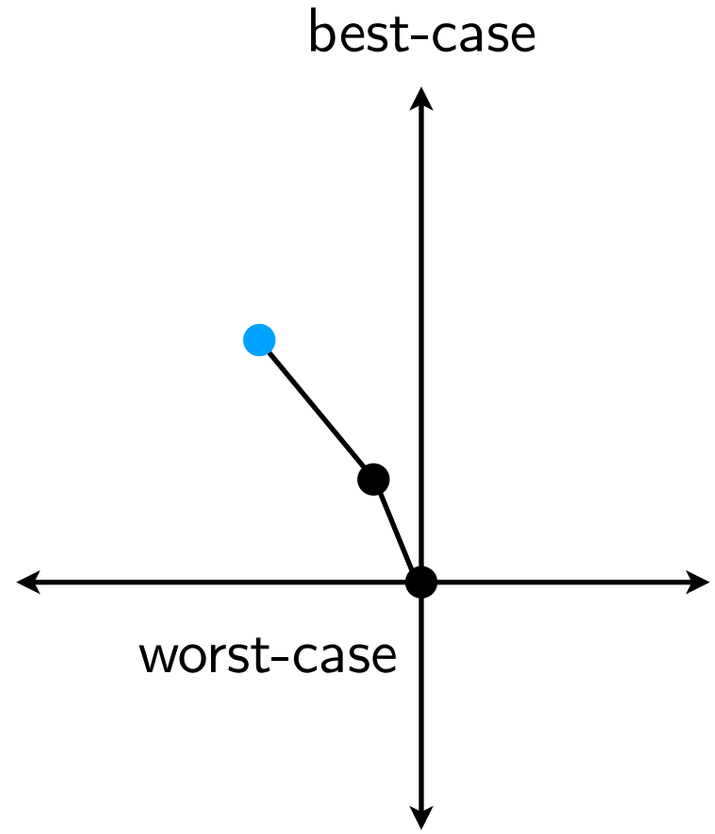
	18-30	**31-65**	**66+**
p_g	0.4	0.2	0.1
$\left\lvert p_g - \frac{1}{2} \right\rvert$	0.1	0.3	0.4



RELY RELY DISTRUST

$\eta \downarrow$

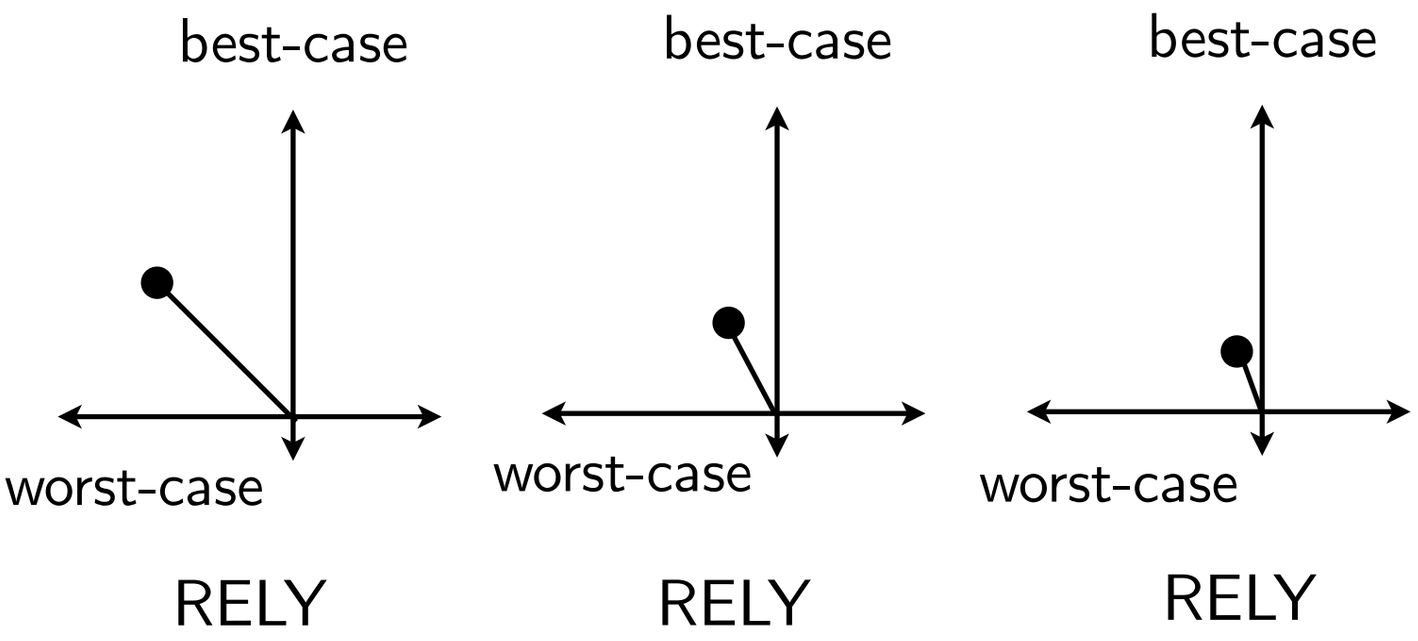
Aggregate Payoffs



(For the figure assume equal population sizes)

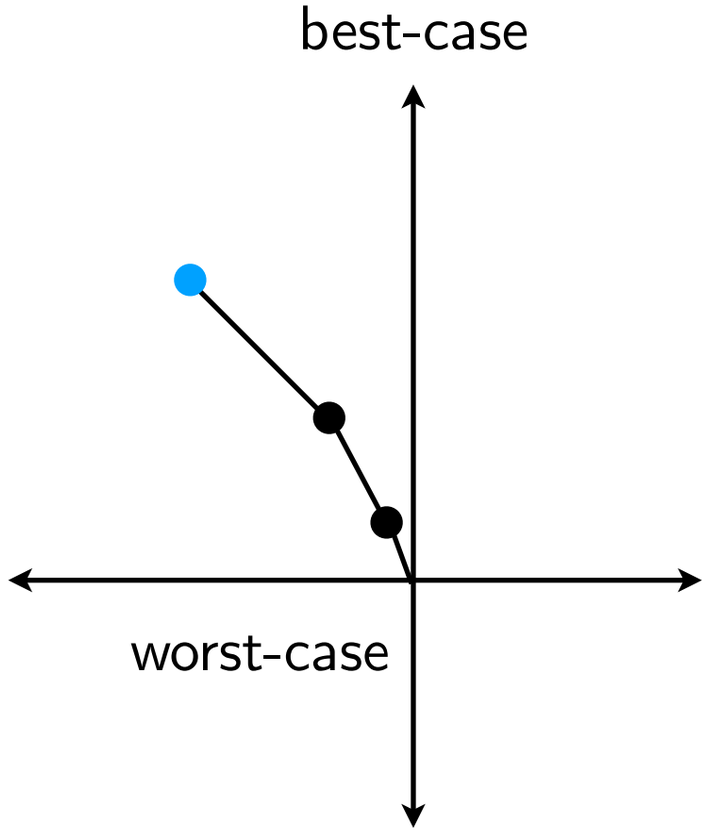
Example 2: Multiple Groups

	18-30	**31-65**	**66+**
p_g	0.4	0.2	0.1
$\left\lvert p_g - \frac{1}{2} \right\rvert$	0.1	0.3	0.4



$\eta \downarrow$

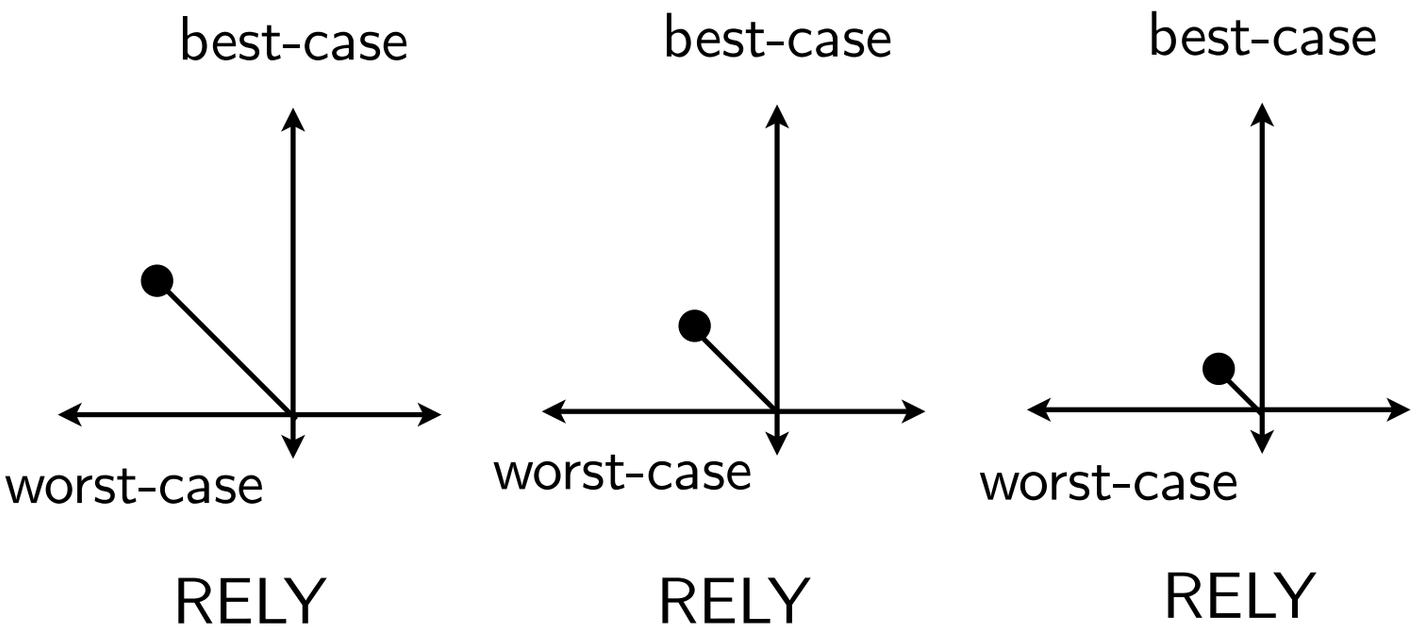
Aggregate Payoffs



(For the figure assume equal population sizes)

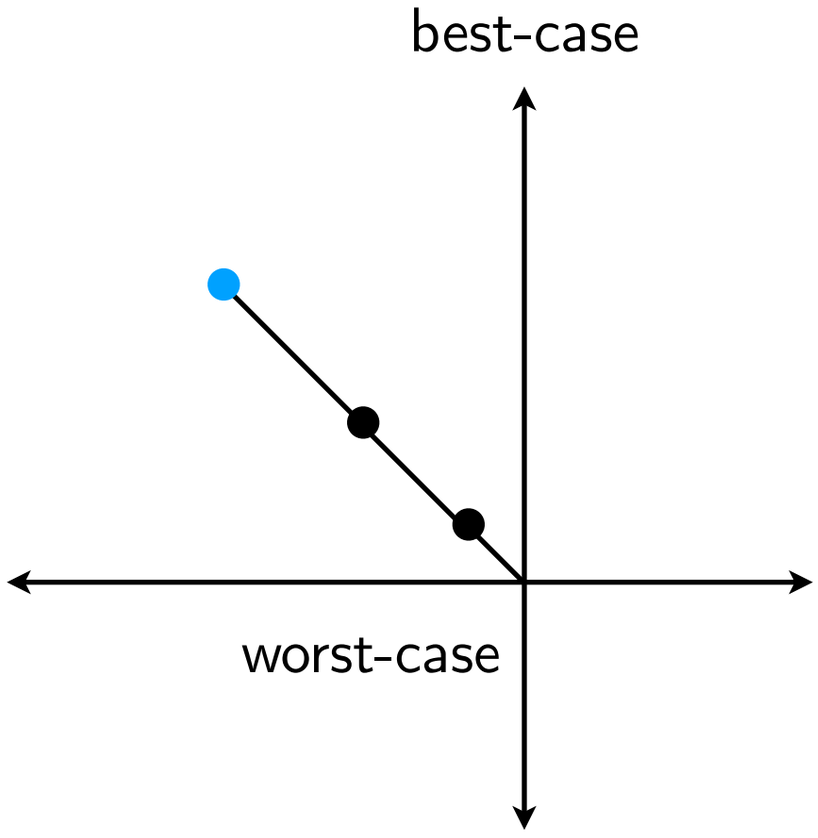
Example 2: Multiple Groups

	18-30	**31-65**	**66+**
p_g	0.4	0.2	0.1
$\left\lvert p_g - \frac{1}{2} \right\rvert$	0.1	0.3	0.4



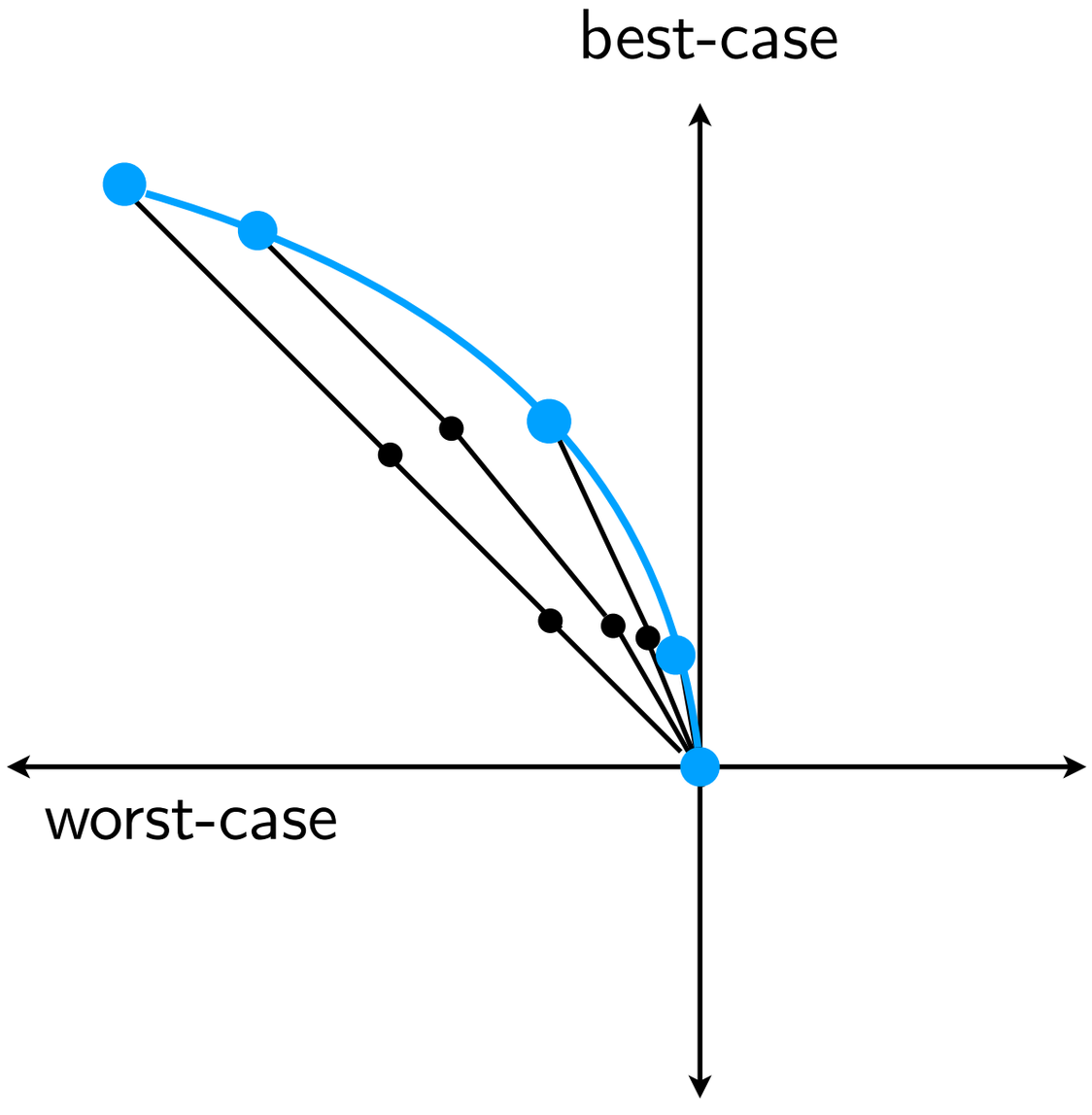
$$\eta = 0$$

Aggregate Payoffs



(For the figure assume equal population sizes)

Example 2: The Risk-Reward Frontier



Takeaways

- Treatment-rate limits alone cannot improve the best-case/worst-case tradeoff
- Input restrictions can improve the tradeoff
  - Don't simply make the AI less informed, but rather control what it can be informed about
  - Allow inputs that identify treatment need but not harm (or the reverse)
- Rely on the AI first in groups where treatment success and harm are most balanced, and expand to more skewed groups as tolerance for risk grows.

Thank You

Benchmark Payoffs

For fixed bounds $(\underline{\tau}, \overline{\tau})$, base rate p and treatment rate r, define $\Pi_p(\tau, r)$ to be the set of all joint distributions π of $(A, Y) \in \{0,1\}^2$ satisfying

- Marginal constraints: $\pi(A = 1) = r$ and $\pi(Y = 1) = p$
- τ-constraints: $\underline{\tau} \leq \pi(Y = 1 \mid A = a) \leq \overline{\tau}$ for each action a

Definition: The best-targeting payoff is

$$b_p(\tau, r) = \sup_{\pi \in \Pi_p(\tau, r)} \mathbb{E}_\pi[u(A, Y)]$$

The worst-targeting payoff is

$$w_p(\tau, r) = \inf_{\pi \in \Pi_p(\tau, r)} \mathbb{E}_\pi[u(A, Y)]$$